

CMP-01 · COMPETENCY FRAMEWORKS

The PCI competency framework — overview

How the Institute states what a certified professional can do,
and what that statement is for

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

The Institute publishes a competency set for each of its three credentials — fourteen for PCL-AI, nineteen for PFL-AI, twenty-four for PML-AI. This document explains what those sets are, how each competency is written, how competency differs from knowledge and from membership standing, and how a candidate, an employer or an assessor uses the framework without misreading it.

— About this document

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The PCI competency framework — overview

In one paragraph. The Institute publishes a competency set for each of its three credentials — fourteen for PCL-AI, nineteen for PFL-AI, twenty-four for PML-AI. This document explains what those sets are, how each competency is written, how competency differs from knowledge and from membership standing, and how a candidate, an employer or an assessor uses the framework without misreading it. After reading it you should be able to take any competency in any of the three sets and say what it would take to demonstrate it, and what would only look like demonstrating it.

Who this is for. Candidates deciding which credential fits the work they actually do; employers writing role specifications, development plans and appraisal criteria; and reviewers and assessors who have to apply the same standard to two different people on two different projects.

— 1. The claim a competency framework has to survive

"Eight years' experience in project controls" is a claim about elapsed time in proximity to a discipline. It is compatible with eight years of running the same monthly report, and with eight years of rebuilding a failing forecast under challenge. Both people write the same line on a CV.

Three moments expose the difference, and all three are expensive:

- **Hiring.** An employer buys the second person and receives the first. The cost surfaces at the first month-end that goes badly.
- **Assessment.** A certification body has to distinguish them from a distance, using a sample of evidence, defensibly enough to withstand an appeal from whoever fails.
- **Dispute.** A number is challenged. The professional either can explain the basis of the number, the method chosen, the alternatives rejected and the sensitivity of the answer to each assumption — or cannot, and the number is replaced by whichever number the room prefers.

A competency framework is the instrument that separates the two people. It does that by stating capability as **decisions a person can make and defend**, not as topics a person has encountered.

— 2. What the framework is made of

Each PCI credential carries a published competency set. The sets live in the platform's certification catalogue, appear on each credential's public page, and are the authoritative statement of what the credential claims.

Credential	Full name	Category	Competencies
PCL-AI	PCI AI Project Controls Leader	Project Controls	14
PFL-AI	PCI AI Project Finance Leader	Project Finance	19
PML-AI	PCI Project Management Leader – AI	Project Management	24

Three consequences follow, and each of them is a rule.

The set is closed. If a capability is not in the set, the credential does not claim it. A holder of PCL-AI has not been assessed on capital structuring; a holder of PFL-AI has not been assessed on agile delivery. This is a feature. A credential that claims everything certifies nothing.

The set is not extensible locally. An employer may map its own framework onto ours, and should. It may not add a fifteenth PCL-AI competency and describe the result as the PCI framework. Changes to a published set are governance decisions, made once, centrally, and published — reviewed on a [\[CONFIRM: published review cycle for the competency sets\]](#) basis.

The counts are a statement of breadth, not of difficulty. Twenty-four is not harder than fourteen. PML-AI spans initiation to closure across predictive, agile and hybrid delivery, so its set is wide. PCL-AI cuts a narrower band deeper — governance, cost, schedule, earned value, forecasting, risk, commercial control and the AI layer over all of them. PFL-AI follows a transaction from appraisal to financial close and then into operations, and picks up the instruments that structure it. All three sit at the same credential level, which is **Leader**. Anyone reading the counts as a ranking has misread the framework in the first thirty seconds.

— 3. The anatomy of a competency statement

Every competency in this series is written in four parts. The structure is uniform so that two assessors reading the same competency reach the same question, and so that a candidate can see what evidence to gather without asking.

Definition. What the competency is, expressed in terms of the decisions it governs. Not "understands cost management" — a definition that cannot fail. Instead: what has to be decided, on what information, against what standard.

By level. What the competency looks like at foundation, practitioner, professional and leader. The scale, its verb grammar and its evidence expectations are defined once, in CMP-02 — Competency levels and how they are evidenced, and are not restated per competency.

Evidence. The named artefacts that would settle the question — a specific document, with a specific author, produced at a specific point in a project. "Relevant documentation" is not evidence; a dated estimate-at-completion note that names the method, the alternatives considered and the reason for the choice is evidence.

The false positive. What is routinely mistaken for the competency. This is in the structure deliberately. The dominant failure mode in competence assessment is not rejecting a capable person; it is accepting an incapable one who has been near capable work and can produce its artefacts. Someone who maintains a risk register is not thereby competent in risk; someone who runs a monthly earned value pack built by someone else is not thereby competent in earned value. Naming the look-alike is what makes the framework usable by an assessor who has fifteen minutes and a folder of documents.

— 4. Competency, knowledge, and the Body of Knowledge

Two different instruments, often confused.

The **Body of Knowledge** states what must be known: thirteen domains and sixty-one knowledge areas, grouped and weighted 40 % project accounting and finance, 40 % project management principles, 20 % AI knowledge and practical approach. PCB-02 — The thirteen domains at a glance and PCB-03 — Why 40/40/20 own that material; this series does not restate it.

The **competency set** states what must be able to be done. Knowledge is necessary and insufficient: a person can recite the four standard estimate-at-completion formulae and still be unable to choose between them in front of a project board, which is the only moment that matters.

The relationship to examination is worth stating precisely, because candidates infer more than is there. The published examination blueprint covers **PCL-AI only**; the domain weighting above is its spine. No examination blueprint exists yet for PFL-AI or PML-AI — no weightings, no item counts. The number of items in any PCI examination is not published anywhere, because it does not yet exist as a decision: it will be confirmed by a formal job-task analysis. The pass mark configured in the platform is 65 %, and the definitive standard will be set by a modified-Angoff standard-setting study. Scoring is criterion-referenced, not graded on a curve. EXB-01 and EXB-05 own the detail.

— 5. Three ladders, and why they are not one ladder

Readers routinely collapse these into a single hierarchy. They are independent, and CMP-02 sets out the relationships in full.

- **Competency level** — foundation, practitioner, professional, leader. A property of a competency held by a person, assessed one competency at a time. Most people sit at different levels across different competencies, and that is normal rather than a defect.
- **Credential level** — the level at which a credential is positioned. All three PCI credentials are recorded at **Leader**. Note that "professional" therefore does double duty in the vocabulary: it is a point on the competency scale and an everyday adjective. In this series the scale word is always written in the level context and never as a synonym for "qualified".
- **Membership grade** — Student Member (SPCI), Associate Member (APCI), Professional Member (MPCI), Fellow (FPCI). A standing within the Institute, not a measure of competence. Professional Member requires an active PCI certification; Fellow is by nomination. **Membership is not certification, and a grade never evidences a competency.**

— 6. Using the framework

As a candidate, before applying. Take the set for the credential you are considering. For each competency, write the single artefact you would hand an assessor, and the month you produced it. The competencies where you cannot name an artefact are your real preparation list. This is a routing exercise, not a score — see CMP-10 §6 on why a self-rating on its own tells you nothing.

As an employer, designing a role. Do not require the whole set at leader level; almost no role needs that, and the specification will filter out everyone you could afford. Select the subset the role actually decides on, set a level per competency, and write the evidence expectation into both the job description and the appraisal so the same words govern hiring and progression. A specification of the form "Forecasting and EAC — professional; Automation — foundation" is checkable. "Strong project controls background" is not.

As a manager, planning development. The useful unit is a gap statement with five fields: the competency, the current level, the target level, the artefact that would demonstrate the move, and the date by which that artefact will exist in the ordinary course of the work. If no such artefact will occur naturally, the development plan is a training wish and will not happen.

As an assessor. CMP-10 — Assessing competence owns evidence types, rubric construction, scoring rules and moderation.

— 7. What this framework is not

It is not a job description, and a competency set copied into a vacancy notice will read as one and mislead. It is not a training syllabus — the sequence in which capabilities are best learnt is not the sequence in which they are listed. It is not an organisational maturity model; it describes people, not functions. It is not a licence to practise, and nothing in it displaces a statutory registration that a jurisdiction requires.

And the status statement, which belongs here rather than in small print: PCI certification is developed with reference to ISO/IEC 17024 personnel-certification principles — the principles governing impartiality, defined competence, separation of training from assessment, and documented assessment decisions. **PCI is not accredited** by any ISO/IEC 17024 accreditation body. The framework claims to be a coherent, published, assessable statement of competence. It does not claim recognition it has not been granted.

— 8. How this goes wrong

The count is read as a ranking. Someone concludes PML-AI is "the hardest" because it has twenty-four competencies, chooses it over the credential matching their work, and is assessed on procurement and benefits realisation they have never touched. Breadth is not depth. Choose on the decisions you make in your job.

A local framework is published under the PCI name. An employer adds three competencies of its own, distributes the result internally as "the PCI competency framework", and two years later a candidate arrives at assessment expecting to be examined on the local additions. Map, do not merge; and label the mapping.

The person is levelled instead of the competency. "She's a practitioner" is not a framework statement. Real profiles are uneven: professional in cost management, foundation in automation. Averaging that into one word destroys exactly the information the framework exists to carry.

Evidence proves attendance, not capability. A training certificate shows a person was in a room. A tool licence shows the organisation bought software. A job title shows someone else's decision about headcount. None of them is evidence of a competency, and an assessor who accepts them has assessed nothing.

The false positive is skipped because it feels negative. It is the most load-bearing part of each competency. Deleting it in a summary — an internal one-pager, a slide — leaves a framework that accredits proximity to good work.

The framework becomes a disciplinary instrument. Used to justify a decision already taken, ratings stop tracking capability within a quarter, everybody learns to rate to the outcome, and the organisation loses its only reliable read on where its real capability gaps are.

Recency is not checked. A person who performed a competency to professional standard six years ago, on a project type that no longer exists in the business, is being credited for a capability that has not been exercised. CMP-02 §6 sets the currency expectation.

— 9. Checklist

Take this into the meeting where the framework is being applied.

- [] Which credential's set are we using — PCL-AI (14), PFL-AI (19) or PML-AI (24)?
- [] Is every competency we are discussing actually in that published set, or has something local crept in?
- [] For each competency in scope, have we set a **level**, not just named the competency?
- [] For each level we have set, can we name the **artefact** that would demonstrate it, its author and its date?
- [] Have we tested each piece of evidence against the competency's **false positive**?
- [] Are we levelling competencies rather than people?
- [] Has anyone in the room conflated the competency scale word "professional" with the membership grade Professional Member (MPCI)?
- [] Does the role specification require leader level on more competencies than the role actually decides?
- [] Is the evidence current, and if not, what would refresh it?
- [] If this is an assessment rather than a development conversation, are we following CMP-10?

— Related

- [CMP-02 – Competency levels and how they are evidenced](#) — the four-level scale, its verb grammar, and the two ladders it must not be confused with
- [CMP-03 – PCL-AI: the fourteen competencies](#) — the project controls set, competency by competency
- [CMP-04 – PFL-AI: the nineteen competencies](#) — the project finance set
- [CMP-05 – PML-AI: the twenty-four competencies](#) — the project management set
- [CMP-10 – Assessing competence – evidence, rubrics and moderation](#) — how a competency claim is actually tested
- [PCB-02 – The thirteen domains at a glance](#) — the knowledge side of the same standard

— Sources and standards

- PCI platform certification catalogue — the seeded competency sets for PCL-AI, PFL-AI and PML-AI ([backend/Data/MultiCert.cs](#)), verified August 2026. These are the authoritative lists; this series explains them and does not extend them.
- ISO/IEC 17024, Conformity assessment — general requirements for bodies operating certification of persons. Referenced for its principles — defined competence, impartiality, separ-

ation of training from assessment, documented decisions — described in our own words. No text is reproduced, and PCI is not accredited to it.

— Status and version

Founding-stage document · Version 1.0 — effective date to be confirmed · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

Competency levels and how they are evidenced

The four-level scale, the evidence that moves a person up it, and the two other ladders it is not



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— Summary

PCI describes competence on a four-point scale — foundation, practitioner, professional and leader — applied one competency at a time. This document defines each level by the decisions it carries rather than by adjectives, sets out the evidence hierarchy that moves a claim from assertion to demonstration, and separates the competency scale from the credential level and from the membership grades, which are different things that share vocabulary.

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Competency levels and how they are evidenced

In one paragraph. PCI describes competence on a four-point scale — foundation, practitioner, professional and leader — applied one competency at a time. This document defines each level by the decisions it carries rather than by adjectives, sets out the evidence hierarchy that moves a claim from assertion to demonstration, and separates the competency scale from the credential level and from the membership grades, which are different things that unfortunately share vocabulary. After reading it you should be able to place a real person on the scale for a real competency and say what evidence would change your placement.

Who this is for. Line managers writing development plans and appraisals, candidates assembling a portfolio of evidence, and assessors who must apply one scale consistently across people they have never met.

— 1. Why levels have to be defined by decisions

Most competency scales fail in the same place. They are written with adjectives — "basic", "good", "strong", "expert" — and adjectives cannot be tested. Two managers rating the same person against "strong cost management" will disagree, discover they disagree only at moderation, and settle it by seniority rather than by evidence.

The scale below is written with **decisions and their consequences**. Each level answers four questions:

1. What does the person decide, and what do they escalate?
2. Who checks the output before it is relied upon?
3. What happens when the situation falls outside procedure?
4. Who carries the error when it turns out to be wrong?

Question four is the sharpest. It is possible to perform sophisticated work at practitioner level; what makes the level professional is that when the number is wrong, the professional is the one who has to explain why — and had the authority to make the choice that produced it.

— 2. The four levels

The scale applies to **one competency at a time**. A person is not "a practitioner"; a person is at practitioner level in planning and scheduling and at professional level in commercial and contract controls. Uneven profiles are the norm and carry the information that matters.

2.1 Foundation

Works within a defined procedure, on a defined part of the work, with the output checked before anyone relies on it. Uses the established method because it is the established method. Recognises

when something looks wrong and escalates rather than resolving it. The organisation carries the error.

The distinguishing test: can this person produce a correct output when the situation is normal?

2.2 Practitioner

Owens a defined scope end to end — a control account, a discipline schedule, a section of a financial model, a work package. The output is relied on without routine correction. Chooses between established methods where the procedure prescribes the choice, and applies the procedure to cases it plainly covers. Escalates the cases it does not.

The distinguishing test: does the work leave this person's hands in a state where the next person builds on it rather than checks it?

2.3 Professional

Decides the method where procedure is silent, and defends the decision to someone who wanted a different answer. Owns the number under challenge: can state the basis, the alternatives rejected, the sensitivity of the result to each assumption, and the point at which they would change their view. Detects and corrects error in other people's work, including work produced by a system. Sets the working standard for a team.

The distinguishing test: when the case is not covered by procedure and the answer is unwelcome, does this person produce a defensible position — and hold it for the right reasons and release it for the right reasons?

2.4 Leader

Sets the rule that others apply, across projects rather than within one. Accountable for the integrity of the system that produces the numbers, including when that system fails — which means owning the failure, diagnosing whether it was an individual error or a design fault, and changing the design. Decides where a method must **not** be used, including where an automated or AI-derived method must not be used, and carries the consequence of that decision. Builds capability deliberately, so that the function survives the departure of any individual.

The distinguishing test: if this person left, would the method survive them — and did they design it so that it would?

2.5 The verb grammar

Level descriptors throughout this series are written with a controlled vocabulary, so that a reader can tell the level from the verb even out of context.

Level	Verbs	Ownership of error
Foundation	assists · prepares · checks against a rule · escalates	The organisation
Practitioner	performs · owns a scope · applies · reconciles	The individual, within scope
Professional	decides · defends · challenges · corrects others	The individual, including the method
Leader	sets · governs · withholds · designs · answers for	The system and its failures

"Understands", "is familiar with", "is aware of" and "has exposure to" appear nowhere in this framework. They describe a mental state that no evidence can reach.

— 3. Three ladders that share vocabulary

This is the section to read twice, because the collisions are real and they mislead people about what they hold.

Ladder	What it measures	Values	Who assigns it
Competency level	Depth of capability in one named competency	foundation · practitioner · professional · leader	Assessment against evidence
Credential level	The positioning of a whole credential	Leader (all three PCI credentials)	The certification scheme
Membership grade	Standing within the Institute	SPCI · APCI · MPCl · FPCI	Membership rules

Credential level. All three credentials — PCL-AI, PFL-AI and PML-AI — are positioned at **Leader** level. That is a statement about the credential, not a claim that every holder is at leader level on every competency in its set. A credential samples a set of competencies and certifies that the holder met the standard on the sample; it does not certify a uniform ceiling.

A recorded inconsistency, stated rather than resolved. The platform database records all three credentials at level Leader. One public page ([certification.html](#)) renders the level as Professional. This series uses **Leader**, in line with the platform record, and treats the competency-scale word professional as an unrelated concept — a point on the scale in §2.3, never a credential position. The discrepancy is genuine and is logged rather than quietly corrected: [\[CONFIRM: single authoritative credential-level wording across platform database and public pages\]](#).

Membership grade. Student Member (SPCI), Associate Member (APCI), Professional Member (MPCI) and Fellow (FPCI) are standings within the Institute. Professional Member requires an active PCI certification; Fellow is by nomination. **A grade is not a competence measure and never evidences a competency.** The traps are specific and both directions occur: reading MPCI as "assessed at professional level" (it is not — it is a membership standing that happens to require a current certification), and reading FPCI as a superlative competence rating (it is a nomination-based standing recognising contribution). A fellow who has not earned a PCI credential holds no PCI certification.

— 4. The evidence hierarchy

A competency claim is worth what its evidence is worth. Evidence types are not interchangeable, and the ordering below is the one an assessor should apply when two sources conflict.

Rank	Evidence type	What it can show	What it cannot
1	An artefact you produced, dated, with your reasoning visible in it	Method choice, judgement under the conditions of the day	That you would repeat it under different conditions
2	An artefact you materially contributed to, with your contribution identifiable	Capability within a defined scope	Sole ownership, or the decisions someone else made
3	A structured professional discussion in which you are questioned on your own artefact	Depth behind the artefact; whether you can defend it	Anything the questioner did not think to probe
4	Direct observation of the work	Behaviour and interaction that artefacts never capture	Anything outside the observed period
5	Attestation by a named person with named standing who saw the work	Corroboration, scope, duration	Detail; and it inherits the attester's judgement
6	Examination performance	Applied knowledge and reasoning against a common standard	Whether the person does it under real pressure
7	Self-report	Where to look next	Capability. See §7 of CMP-10

Two practical rules follow.

Reasoning must be visible in the artefact, not reconstructed afterwards. A forecast that records the method used, the alternatives rejected and the assumption the answer hangs on is rank-1 evidence. The same forecast without that record is a number, and a number can be produced by a spreadsheet someone else built. The habit of writing the basis into the artefact at the time is the single cheapest thing a candidate can do to make their own competence provable later.

Contribution must be identifiable at the time, not asserted later. "I built the risk model" is uncheckable two years on. A model with a version history, a named owner per section, and an issue log carrying your name against specific resolutions is checkable.

— 5. Evidence by level

The type of evidence changes as the level rises, because the thing being demonstrated changes.

Level	What must be shown	Representative evidence
Foundation	Correct output under normal conditions, and escalation of the abnormal	Completed work to procedure; a raised query that turned out to be right; a checked-and-accepted deliverable
Practitioner	An owned scope, relied on without correction	A control account or model section you own; a month-end pack you produce; a reconciliation you sign
Professional	A defended decision where procedure was silent	A dated method-selection note; a forecast you revised upward against pressure, with the basis; a correction you made to a superior's number, and the outcome
Leader	A rule others apply, and accountability when it fails	A procedure or standard you authored and that is in use; a decision to withhold a method or tool, with reasoning; a post-incident change to the system after a failure you owned

Note the shape of professional-level evidence: it usually documents **friction**. If a candidate's entire portfolio is smooth — every forecast accepted, every position unchallenged — the portfolio is showing practitioner-level work performed pleasantly, or a professional whose contemporaneous record is missing. CMP-09 — Behavioural and leadership competencies covers what that friction looks like when handled well.

— 6. Currency: when evidence expires

Competence decays, and it decays unevenly. Judgement about contract mechanics survives a decade; fluency in a reporting toolchain or an AI-assisted method does not survive two years of disuse.

The Institute's working position:

- Evidence more than **three years** old is treated as historical. It supports a claim; it does not carry one alone. Three years is chosen deliberately to align with the certification cycle.
- Evidence for any competency naming AI, automation or digital delivery is treated as **fast-decaying**: the method changes, and unexercised familiarity is not capability. Recertification therefore carries a mandatory AI-currency component, and the platform's continuing professional development category satisfying it is named exactly that. CER-08 owns the recertification rules, including the binding continuing professional development requirement, which is not yet published [CONFIRM: binding CPD hours per three-year cycle — the student portal shows a 30-hour target, which is not a rule].
- A competency exercised on one project type only is evidenced for that context. Say so rather than generalising it; an assessor will find the boundary anyway, and finding it themselves costs the candidate credibility they did not need to lose.

— 7. How this goes wrong

Levels are assigned by seniority. A manager is rated professional because they are a manager. The rating then becomes unfalsifiable, because nobody is going to rate their own manager down, and the scale stops carrying information within a cycle.

One level is assigned to a whole person. The single most common misuse. It destroys the uneven profile, which is the actionable part: the person who is professional in earned value and foundation in automation has an obvious, cheap development plan, and "practitioner overall" hides it.

"Professional" is read as a rank rather than a scale point. Someone at professional level in three competencies believes they hold a professional standing, or believes they are equivalent to a Professional Member (MPCI). They are different ladders (§3). Where both appear in one document, label both.

Leader level is claimed for scope rather than for rule-setting. Managing thirty people is a span of control. Leader level requires that a method exists because you decided it should, that others apply it, and that you answered for it when it failed. Headcount is not evidence.

Evidence is gathered retrospectively at application time. The candidate reconstructs reasoning from memory, and the reconstruction is uniformly favourable, because nobody remembers the assumption they failed to test. This is not dishonesty; it is how memory works. The counter is contemporaneous recording, which costs a paragraph at the time.

A training certificate is filed as evidence of a competency. It evidences attendance. It becomes evidence of competency only when accompanied by work produced afterwards that the training explains.

Currency is never tested. A portfolio is accepted whole, including a demonstration from a project that closed six years ago, on a delivery model the organisation abandoned. Ask when, not just whether.

The scale is imported into pay banding unchanged. The moment a level determines a salary band, every rating conversation becomes a negotiation, evidence standards fall, and the framework's diagnostic value is gone. Use it for development and assessment; if it must inform reward, put a moderation step (CMP-10 §5) between the rating and the money.

— 8. Checklist

For a levelling conversation about a real person:

- [] Are we levelling **one competency at a time**, by name?
 - [] For each level claimed, which of the four questions in §1 have we actually answered — decision, check, procedure-silence, ownership of error?
 - [] Is the evidence rank 1 or 2 in §4, or are we leaning on rank 5–7?
 - [] Is the reasoning visible **inside** the artefact, or reconstructed in the conversation?
 - [] For a professional-level claim, can we point to a decision made where procedure was silent, and to how it was defended?
 - [] For a leader-level claim, can we name the rule, the people applying it, and a failure that was owned?
 - [] How old is the evidence, and does the competency decay fast (§6)?
 - [] Has anyone in the room used "professional" to mean the membership grade, or "leader" to mean seniority?
 - [] Would a second assessor, given the same folder and this document, reach the same level?
-

— Related

- [CMP-01 – The PCI competency framework – overview](#) — what the sets are and how each competency is written
 - [CMP-03 – PCL-AI: the fourteen competencies](#) — the scale applied competency by competency
 - [CMP-09 – Behavioural and leadership competencies](#) — the behaviours that professional-level evidence usually documents
 - [CMP-10 – Assessing competence – evidence, rubrics and moderation](#) — rubric wording, scoring rules and moderation
 - [CAR-07 – Building a portfolio of evidence](#) — how a candidate assembles and dates the artefacts
 - [CER-08 – Recertification, CPD and the AI-currency requirement](#) — the currency rules for a held credential
-

— Sources and standards

- PCI platform certification catalogue (<backend/Data/MultiCert.cs>), verified August 2026 — the credential `level` field is recorded as Leader for PCL-AI, PFL-AI and PML-AI. The divergent Professional rendering on the public certification page is recorded in §3 as an open discrepancy.
- Membership grades and their requirements as published by the Institute: SPCI, APCI, MPCCI (requires an active PCI certification), FPCI (by nomination).
- ISO/IEC 17024, Conformity assessment — general requirements for bodies operating certification of persons. Referenced for the principle that competence must be defined before it is assessed and that assessment decisions must be documented and reviewable. Described in our own words; no text reproduced; PCI is not accredited to it.

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CMP-03 · COMPETENCY FRAMEWORKS

PCL-AI: the fourteen competencies

Each competency of the PCI AI Project Controls Leader —
defined, levelled, evidenced, and separated from its look-alike

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

The PCI AI Project Controls Leader credential publishes fourteen competencies. This document takes each one in turn and states what it is in terms of decisions, what it looks like at each of the four levels, the named artefacts that evidence it, and the common false positive — the thing that is routinely mistaken for the competency and is not it.

— About this document

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PCL-AI: the fourteen competencies

from its look-alike.

In one paragraph. The PCI AI Project Controls Leader credential publishes fourteen competencies. This document takes each one in turn and states what it is in terms of decisions, what it looks like at each of the four levels, the named artefacts that evidence it, and the common false positive — the thing that is routinely mistaken for the competency and is not it. It is the working reference for anyone assembling evidence, writing a role specification, or assessing a claim against the PCL-AI standard.

Who this is for. Candidates for PCL-AI; project controls managers and heads of function designing roles and development plans; assessors applying the standard to evidence they did not collect.

— 1. How to read this document

The fourteen competencies below are the published PCL-AI set, in the published order. They are not reordered, renamed, merged or extended here. Where two of them appear to overlap — and several do, by design — the boundary is stated explicitly, because the boundary is where assessment decisions are actually made.

Each entry has four parts, per CMP-01 §3: **definition** (the decisions the competency governs), **by level** (foundation, practitioner, professional, leader — the scale is defined in CMP-02 §2 and is not restated here), **evidence** (named artefacts), and **the false positive** (what looks like it and is not).

Three clusters are treated in greater depth elsewhere and are cross-referenced rather than repeated: cost and commercial in CMP-06, planning, scheduling and risk in CMP-07, and the data, digital and AI group in CMP-08.

One boundary of scope is worth stating up front. The published examination blueprint for PCL-AI carries the Body of Knowledge domain weighting; it does not publish a mapping from these fourteen competencies to those domains [\[CONFIRM: published competency-to-domain mapping for the PCL-AI examination blueprint\]](#). The competency set states what the credential claims; the blueprint states how the examination samples knowledge. They are complementary and neither is derived from the other.

— 2. Governance, cost, schedule and value

2.1 Project controls governance

Definition. Deciding how the controls function itself operates: what is baselined and by whom, which changes require which authority, the calendar on which data is cut and reported, the segregation between the person who forecasts and the person who approves, and what happens when a

rule is broken. Governance is the competency that makes the other thirteen mean the same thing twice.

By level. Foundation — follows the controls procedure and knows which approvals a change requires. Practitioner — operates the governance calendar for a scope: runs the cut-off, maintains the change log, prepares change papers to the required authority. Professional — designs the governance for a project: sets thresholds, defines the delegation of authority in controls terms, and refuses to process a change that has not been authorised, including when the requester is senior. Leader — owns the controls governance framework across a portfolio, decides where projects may deviate from it, and rebuilds it after a failure rather than reissuing the same document with a stern covering note.

Evidence. A project controls execution plan you authored; a change authority matrix in force with your name against it; a cut-off calendar with adherence recorded; a documented refusal or escalation of an unauthorised change; a governance review finding you raised and the change that followed.

The false positive. Owning the procedure document. A well-written procedure that nobody follows is evidence of authorship, not governance. Ask what has been stopped by the governance in the last six months; if the answer is nothing, the governance is decoration.

2.2 Cost management

Definition. Controlling the flow from estimate to budget to commitment to accrual to actual cost to forecast, keeping each step reconcilable to the last, and knowing at any date what has been spent, what has been committed but not spent, and what has been incurred but not yet invoiced. Depth in CMP-06.

By level. Foundation — codes and processes cost transactions against the cost breakdown structure, and queries a misallocation. Practitioner — owns control accounts: maintains commitments, raises and substantiates accruals, reconciles to the ledger, explains a variance in terms of its driver rather than its size. Professional — sets the cost breakdown structure and the code of accounts for a project, decides accrual policy at the boundary cases, and defends a cost position to a finance function that computes it differently. Leader — governs the cost control environment across projects: the structures, the cut-off discipline, the interface with corporate finance, and the standard of substantiation required.

Evidence. A cost breakdown structure and code of accounts you designed; a signed month-end accrual pack with your substantiation; a documented reconciliation between the project cost report and the general ledger, including the explained differences; a variance analysis that identified a driver later confirmed.

The false positive. Cost reporting. Producing an accurate report of what a system contains is administration. Cost management is the competency that determines whether what the system contains is right — that the accrual reflects work done rather than invoices received, that a commitment is a commitment and not an intention.

2.3 Planning and scheduling

Definition. Building and owning a schedule whose logic represents how the work will actually be done, maintaining it as reality diverges from it, and being able to state what is driving the completion date and what would change it. Depth in CMP-07.

By level. Foundation — updates progress against an existing schedule and flags logic that has become impossible. Practitioner — owns a discipline or area schedule: builds the network, applies calendars and resources, runs the update cycle, and explains a movement in the critical path in terms of the activities that caused it. Professional — decides the schedule architecture for a project — level structure, coding, calendar strategy, how contractor schedules integrate — and defends a completion forecast against a delivery team that wants an earlier date. Leader — sets scheduling standards across projects, decides where constraint-driven planning is legitimate and where it is a disguised instruction, and owns schedule credibility with the client or board.

Evidence. A schedule you built, with its basis-of-schedule document; a documented critical path analysis before and after an event; a schedule quality review you performed on someone else's programme with the findings and the resolution; a completion forecast you held that was later borne out, with the contemporaneous note.

The false positive. Tool fluency. Rapid competence in scheduling software with no view on whether the logic is credible is data entry at speed. The test is not what the software shows but whether the person can say why the current critical path is the critical path — and whether they have ever concluded that it is not.

2.4 Earned value management

Definition. Operating the measurement system that relates work performed to work planned and to cost incurred: the scope-to-budget structure, the rules of credit that decide how progress is claimed, the integrity of the resulting earned value, and the variance analysis that follows from it. Expanded in CMP-06 §3.

By level. Foundation — applies rules of credit correctly and calculates the standard indices. Practitioner — owns the performance measurement baseline for a scope, applies and defends rules of credit at the awkward boundaries (procured but not installed, fabricated off site, engineering substantially complete), and produces variance analysis with causes attached. Professional — designs the earned value system for a project: work package sizing, technique selection per work type, the treatment of level-of-effort scope, and the rules that prevent progress being claimed for work that has not happened. Detects and corrects earned value that has drifted from physical reality. Leader — sets the standard for earned value across projects and decides where it should not be used at all, because the scope will not support honest measurement.

Evidence. A rules-of-credit sheet you wrote and applied; a performance measurement baseline with your work-package structure; a documented instance where you rejected a progress claim and the reason; variance analyses with root causes that were subsequently confirmed by events.

The false positive. Producing indices. A cost performance index calculated to three decimal places from a system nobody has validated is arithmetic. Earned value competence is the ability to say whether the earned value is real — which usually means having walked the work, or having found the cases where the claim and the physical position disagreed.

2.5 Forecasting and EAC

Definition. Producing a forward view of final cost and date — the estimate at completion (EAC) and the forecast completion — choosing the method deliberately, stating the assumptions the answer depends on, and defending the result when it is unwelcome. Depth in CMP-06 §4.

By level. Foundation — calculates an estimate at completion using the prescribed formula and identifies where the inputs look wrong. Practitioner — produces the periodic forecast for a scope,

selecting from established methods according to procedure, and explains the movement since last period. Professional — decides which method fits the situation where procedure is silent — performance-index based, remaining-work re-estimate, or a hybrid — states the range and its drivers, and holds the forecast when it is challenged, changing it only on evidence. Leader — governs forecasting across a portfolio: the method standard, the review cadence, the treatment of undisclosed exposure, and the escalation route when a forecast is being suppressed.

Evidence. Dated forecast notes recording method, alternatives rejected and key assumptions; a forecast you revised upward before external events forced it, with the record; a back-test of your own forecasts against outturn; a range or scenario forecast with the drivers of the spread named.

The false positive. A single number produced by the tool's default formula. The competency lives in the method selection and the assumption statement, not in the arithmetic. A person who cannot say what would have to be true for their forecast to be right does not hold this competency at professional level, however sophisticated the model behind it.

2.6 Performance measurement

Definition. Deciding what to measure so that the measures describe the project rather than flatter it: physical progress against claimed progress, productivity, quality and rework, interface and dependency health, and the leading indicators that move before the lagging ones. Broader than earned value, which is one measurement system among several.

By level. Foundation — collects measurement data accurately and to the defined cadence. Practitioner — operates the measurement set for a scope, validates data at source, and notices when a measure has stopped tracking reality. Professional — designs the measurement set for a project: chooses the measures, defines each one operationally (what counts, who counts it, at what frequency, with what tolerance), and removes measures that have become performative. Leader — sets what the organisation measures, and accepts the consequence that people optimise what is measured; designs measure sets that survive that pressure.

Evidence. A measurement definition sheet you authored, with operational definitions per measure; an instance where you retired or replaced a measure and why; a documented case where a leading indicator you introduced anticipated an outcome; a data-quality audit at source with the corrections made.

The false positive. Dashboard richness. Twenty-four indicators, six of which are actionable and none of which is defined operationally, is measurement theatre. The competency test is whether the person can name a decision that a measure changed.

2.7 Project risk

Definition. Identifying what could change the outcome, quantifying it where quantification is meaningful, assigning ownership that is real, and connecting the analysis to the contingency and the decisions it should inform. Depth in CMP-07 §6.

By level. Foundation — records risks accurately, maintains the register, chases owners for updates. Practitioner — facilitates risk identification for a scope, assesses probability and impact with a defined scale, tracks mitigation actions to completion and closes risks that have expired. Professional — decides the analysis approach — qualitative, quantitative cost risk analysis, quantitative schedule risk analysis (QSRA) — designs the modelling, interprets the output honestly including its sensitivity to correlation assumptions, and sets contingency on that basis. Challenges risk scores that are political rather than analytical. Leader — governs risk across projects, owns the relation-

ship between risk analysis and contingency policy, and decides when a risk position must be escalated regardless of who is uncomfortable.

Evidence. A quantitative risk analysis you ran, with the input basis, correlation assumptions and the contingency recommendation that followed; a register entry you closed as expired with reasoning; a risk you escalated that materialised; a documented challenge to an impact assessment you judged understated.

The false positive. Register maintenance. A tidy register with current dates and complete fields demonstrates administration. Risk competence shows up in whether the register contains the risks that actually matter — and specifically whether it contains the one nobody wants written down.

2.8 Commercial and contract controls

Definition. Controlling the commercial position: what has been instructed, what has been valued, what has been agreed, what is claimed and unresolved, and what exposure sits outside the current forecast. Includes change and variation control, the substantiation of entitlement, and the interface with the contract's notice and time-bar mechanics. Depth in CMP-06 §5.

By level. Foundation — logs instructions and variations, and knows which contract clause governs a notice. Practitioner — owns the change and variation register for a scope, values changes to procedure, maintains the record required to substantiate an entitlement, and raises notices within their period. Professional — decides the commercial position on contested change: what to price, what to reserve, what to disclose as exposure rather than book as forecast; assembles the factual record supporting an extension of time (EOT) or additional cost claim; and states the commercial risk to the project board in terms they can act on. Leader — sets the standard for commercial control across projects, including the boundary between the controls function and the commercial or legal function, and owns the organisation's exposure reporting.

Evidence. A change control log you owned, with the audit trail from instruction to agreed value; a contemporaneous records regime you set up and the claim it later supported; an exposure report distinguishing forecast from unquantified exposure; a notice issued on time under a time-bar provision.

The false positive. Contract knowledge. Being able to cite the clause is necessary and not sufficient. The competency is in the discipline of the record — whether the daily allocation sheets, the delay records and the correspondence exist and are organised well enough that entitlement can be demonstrated eighteen months later by someone who was not there.

— 3. The data, digital and AI group

These six competencies form a chain, not a set of alternatives: data and analytics produce a proposal, the controls design decides where that proposal enters the work, reporting and automation carry it, responsible AI governs it, and human validation makes a person answerable for it. CMP-08 treats the chain in depth; the entries here define each link and its boundary.

3.1 Predictive analytics

Definition. Using quantitative technique on project data to say something about what has not happened yet — trend extrapolation, regression against drivers, benchmark comparison, statistical modelling of outcome ranges — and knowing the conditions under which each technique is valid.

By level. Foundation — reads and correctly interprets an analytical output produced by someone else, including its stated uncertainty. Practitioner — applies established analytical techniques to project data, prepares and cleans the data, and reports the result with its confidence and its limitations. Professional — selects the technique for the question, tests the assumptions that make it valid, holds out data to back-test, and reports the error characteristics of their own model rather than only its central answer. Leader — decides what the organisation will and will not predict analytically, sets the data standard that makes prediction possible, and owns the consequence when a model is used outside its range.

Evidence. An analysis with its data basis, method rationale and validation results; a back-test of your own predictions against actuals; a documented decision that the data would not support the analysis requested, and what you did instead.

The false positive. Curve fitting to date. A trend line through eight months of cost data with a high coefficient of determination is a description of the past, not a prediction, and the coefficient says nothing about whether the relationship will hold. The competency test is whether the person has ever reported that their own model was wrong, and by how much.

3.2 AI-enabled project controls

Definition. Designing how the controls function actually uses artificial intelligence: which tasks are delegated to a model, what the model receives, what it returns, where its output enters the process, what check stands between the output and reliance, and what happens when the model is unavailable or wrong.

By level. Foundation — uses approved AI-assisted tools within the defined process and knows which outputs must be checked before use. Practitioner — operates an AI-assisted workflow end to end, prepares inputs to the required standard, and recognises output that is implausible for the project. Professional — designs the workflow: selects the task, specifies the input and output contract, defines the check, and decides which tasks are not delegated at all. Can state the failure mode of each delegated task. Leader — sets the controls function's AI operating model, evaluates and admits tools, decides where the function will not use AI, and owns the outcome of those decisions.

Evidence. A process map showing AI-assisted steps with their input, output and check points; a tool evaluation you conducted with the criteria and the decision; a task you removed from an AI-assisted workflow, and why; an incident review after an AI-derived output reached a report.

The false positive. Tool adoption. Buying and using a capable product is procurement plus practice. The competency is the design — knowing precisely where in the month-end the model's output enters, and what would catch it if it were confidently wrong. AIG-11 covers tool evaluation in depth.

3.3 Digital reporting

Definition. Building the path from source data to the decision it informs so that the number in the report can be traced to its origin, is produced the same way every period, and is available in time to change something.

By level. Foundation — produces reports from the established pipeline and identifies data that looks wrong. Practitioner — owns a reporting product: manages its data sources, its refresh cycle and its definitions, and resolves discrepancies between it and the underlying systems. Professional — designs the reporting architecture for a project: source of truth per data item, lineage, definition

ownership, the audience-by-audience content decision, and the deliberate removal of what nobody uses. Leader — governs reporting across the portfolio, including the hard part: retiring reports, and enforcing one definition of a term across functions that each prefer their own.

Evidence. A reporting architecture or data-lineage document you produced; a reconciliation regime between the report and its sources; a definition dispute you resolved and the decision record; a report you decommissioned.

The false positive. Visual quality. A clear, well-designed dashboard drawing on an unreconciled extract is an attractive route to a wrong decision. The competency is in the lineage and the definitions, not the rendering. BPG-15 covers the visualisation craft.

3.4 Automation

Definition. Removing manual steps from the controls cycle without removing the controls: deciding what to automate, designing the automation so its failures are visible rather than silent, and maintaining it as the surrounding systems change.

By level. Foundation — runs automated routines and reports when one fails. Practitioner — builds and maintains automation for a defined task, with error handling and a reconciliation that proves the output. Professional — decides what should be automated and what should not, on the basis of frequency, stability and the cost of a silent error; builds in the break-detection that makes failure loud; and documents it so it survives the author. Leader — sets the automation standard across the function, including ownership, change control and the requirement that no automation is a single-person dependency.

Evidence. An automated process with its documentation, error handling and reconciliation control; a break you designed for and that later caught a real failure; a decision not to automate a task, with reasoning; a handover in which someone else took over your automation without you.

The false positive. Volume of scripts. Automation that works only while its author is present, or that fails silently and produces a plausible wrong number, has increased risk while appearing to increase efficiency. The question that separates the levels is: when this breaks, how will anyone know?

3.5 Responsible AI

Definition. Governing the use of AI so that its outputs are explainable, its data use is lawful and proportionate, its known biases are considered before reliance, its use is disclosed where a reader would reasonably want to know, and there are named cases where it is not used at all.

By level. Foundation — follows the organisation's AI use policy, including on confidential project data. Practitioner — applies the policy in practice: knows what may be entered into which tool, discloses AI assistance where required, and raises outputs that look biased or unexplainable. Professional — decides the responsible-use position for a project: what may be delegated, what must be disclosed, what data may be used, and where explainability is a precondition of reliance rather than a nicety. Leader — owns the AI governance framework, including procurement criteria, incident handling, and the decision to prohibit a use case the organisation would find commercially convenient.

Evidence. An AI use policy or project-level position you authored; a disclosure convention you introduced into reporting; a documented refusal of a use case, with reasoning; an incident review after an AI-related error, with the control change that followed.

The false positive. Policy possession. Having a policy, or having attended AI ethics training, is not the competency. The competency shows in a decision that cost something — a tool not adopted, a use case declined, a disclosure made that invited an awkward question. AIG-08 and AIG-09 own the underlying framework.

3.6 Human validation

Definition. The specific, personal act of checking an AI-derived or automated output before it is relied on, and taking ownership of it as your own work. Includes designing the check so it is capable of detecting the error that matters, performing it, recording it, and being answerable for what you signed.

By level. Foundation — performs the prescribed check and escalates a failure rather than passing it on. Practitioner — validates outputs within their scope against independent sources, and records what was checked and what was found. Professional — designs the validation for a workflow: what an independent check is for this output, what sample size or coverage is enough, which errors the check would miss, and what evidence the check leaves behind. Rejects output that cannot be validated. Leader — sets the validation standard for the function, ensures that validation cannot be discharged by a tick, and owns the failures of validated output as failures of the standard rather than of the individual.

Evidence. A validation protocol you designed, naming the independent source and the tolerance; signed validation records with findings, including findings that stopped a release; a rejection of an unvalidatable output and what replaced it; a post-incident revision of a check that had missed something.

The false positive. The review tick. A signature confirming that output "has been reviewed", with no statement of what was checked against what, is a record of a signature. The house position is that AI proposes and the professional disposes; that is only true where the disposal is a real, documented, independent check performed by someone who could have caught the error.

— 4. Where the boundaries actually lie

Four pairs are confused often enough to be worth stating flatly.

Earned value management and performance measurement. Earned value is one measurement system, concerned with work performed against plan and cost. Performance measurement is the decision about what the project measures at all, including everything earned value cannot see — productivity, rework, interface health, data quality.

Predictive analytics and AI-enabled project controls. Predictive analytics is a class of technique applied to data. AI-enabled project controls is an operating-model competency: where any such technique enters the process, under what control. A person can be strong in one and absent in the other.

Automation and digital reporting. Automation removes manual steps; digital reporting establishes what the number means and where it came from. Automating an unreconciled report produces wrong numbers faster.

Responsible AI and human validation. Responsible AI is the governing position — what may be used, by whom, disclosed how. Human validation is the individual act at the point of reliance. An

organisation can have exemplary policy and no validation, and the failures will look like model failures when they are control failures.

— 5. How this goes wrong

The list is treated as fourteen boxes to tick, all at leader level. Nobody has that profile, and a candidate who claims it is telling the assessor something unintended. Real evidence portfolios are uneven; say which competencies are your depth and which are your working knowledge.

Overlapping competencies are evidenced with the same artefact four times. One month-end pack offered as evidence for cost management, earned value, performance measurement and digital reporting. It may legitimately support two, if the artefact shows different decisions. Reusing it across four means the candidate has not distinguished the competencies, and the assessor will notice.

The AI competencies are evidenced by tools rather than decisions. A list of platforms used answers a question nobody asked. Every one of the six data and AI competencies is evidenced by a decision: where the output enters, what check stands between it and reliance, what was declined.

The false positive is defended rather than addressed. A candidate whose risk evidence is register maintenance argues that maintaining the register is important. It is. It is also not the competency, and arguing the point burns the assessor's goodwill on the competency where the candidate is weakest.

Governance is evidenced by documents nobody follows. The quickest test an assessor applies to a governance claim is to ask what the governance has prevented recently. A blank answer converts a claimed leader-level competency into evidence of authorship.

The set is quietly reordered or renamed to fit an in-house framework. It then no longer maps to the credential, and a candidate prepares against a set that does not exist. Map onto the published set and keep the published names.

Recency is not stated. A strong demonstration from a project that closed five years ago is offered without a date, and the omission reads as concealment even when it is not. Date every artefact, and say plainly where a competency has not been exercised lately.

— 6. Checklist

For a candidate assembling evidence, or an assessor reviewing it:

- [] Have all fourteen competencies been addressed, including the ones where the honest answer is "practitioner, and here is why"?
- [] Does each competency have at least one **dated artefact**, with an identifiable contribution?
- [] Is any single artefact doing duty for more than two competencies?
- [] For each professional-level claim, does the evidence show a decision made where procedure was silent?
- [] For each leader-level claim, is there a rule others applied, and a failure that was owned?
- [] Has each competency been tested against its **false positive** in §2 and §3?
- [] Are the six data and AI competencies evidenced by decisions rather than by tool names?
- [] Is human validation evidenced by a protocol and findings, rather than by a review signature?

- [] Is the boundary between overlapping competencies (§4) reflected in the evidence chosen?
- [] Is every artefact dated, and is anything older than three years flagged as historical?

— Related

- [CMP-01 – The PCI competency framework – overview](#) — what the set is for and how each entry is structured
- [CMP-02 – Competency levels and how they are evidenced](#) — the four-level scale used throughout this document
- [CMP-06 – Cost and commercial competencies in depth](#) — cost management, earned value, forecasting and commercial control at working depth
- [CMP-07 – Planning, scheduling and risk competencies in depth](#) — schedule and risk competence, including quantitative analysis
- [CMP-08 – Data, digital and AI competencies in depth](#) — the six-competency chain from analytics to human validation
- [CMP-10 – Assessing competence – evidence, rubrics and moderation](#) — how these claims are actually tested

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the PCL-AI seeded competency set, fourteen items, reproduced here in the published order and wording.
- PCI Body of Knowledge domains 4, 5, 6, 7, 10, 12 and 13 ([docs/bok/](#)) — the knowledge base underlying these competencies. Cited by domain; not reproduced.

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CMP-04 · COMPETENCY FRAMEWORKS

PFL-AI: the nineteen competencies

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VERSION 1.0 · DRAFT
2026-08-04

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PFL-AI: the nineteen competencies

its look-alike.

In one paragraph. The PCI AI Project Finance Leader credential publishes nineteen competencies. This document takes each one in turn and states what it is in terms of decisions, what it looks like at each of the four levels, the named artefacts that evidence it, and the common false positive — the thing that is routinely mistaken for the competency and is not it. It is the working reference for anyone assembling evidence, specifying a project finance role, or assessing a claim against the PFL-AI standard.

Who this is for. Candidates for PFL-AI; heads of project finance, transaction leads and lending teams designing roles and development plans; assessors applying the standard to evidence they did not collect.

— 1. How to read this document

The nineteen competencies below are the published PFL-AI set, in the published order, unaltered. Each entry follows the structure set out in CMP-01 §3: **definition** (the decisions the competency governs), **by level** (the four-level scale defined in CMP-02 §2 and not restated here), **evidence** (named artefacts), and **the false positive**.

One scope statement first, because candidates infer more than exists. **No examination blueprint has been published for PFL-AI** — no domain weightings, no item counts [\[CONFIRM: PFL-AI examination blueprint — weightings and item counts\]](#). The competency set below is the authoritative published statement of what the credential claims; it is not a syllabus with a weighting, and it should not be read as one. Item counts for any PCI examination await a formal job-task analysis, and the definitive pass mark will be set by a modified-Angoff standard-setting study; the platform's configured mark is 65 % and scoring is criterion-referenced, not graded on a curve.

Acronyms used throughout are expanded on first appearance: special purpose vehicle (SPV), cash flow available for debt service (CFADS), debt service cover ratio (DSCR), loan life cover ratio (LLCR), project life cover ratio (PLCR), public-private partnership (PPP), environmental and social (E&S), internal rate of return (IRR), net present value (NPV), weighted average cost of capital (WACC).

— 2. Foundations and appraisal

2.1 Project finance

Definition. Financing an asset on the strength of its own cash flows, in a ring-fenced vehicle, with recourse to sponsors limited by agreement rather than by sentiment. The competency is the ability

to place a transaction correctly on the recourse spectrum and to work out who bears which risk once the structure is built.

By level. Foundation — describes the structure of a financing accurately: the SPV, the sponsors, the lenders, the principal contracts. Practitioner — works inside a live transaction, tracing how a particular risk is allocated through the contract suite and into the financing documents. Professional — decides whether a project should be financed on a limited-recourse basis at all, and identifies the risks that the structure has failed to allocate to anyone. Leader — sets a firm's or a lender's approach to structuring, and declines transactions whose risk allocation is unsound however attractive the returns.

Evidence. A structure memorandum you authored; a risk allocation matrix tracing each material risk to the party bearing it and the document that puts it there; a documented recommendation against a structure, with reasoning and outcome.

The false positive. Vocabulary. Fluency in the terminology of project finance is acquired quickly and proves nothing. The diagnostic question is the unglamorous one: which risk in this structure is currently unallocated, and who will actually pay if it happens? Structures fail at the risks nobody was assigned.

2.2 Infrastructure finance

Definition. Applying project finance to infrastructure assets specifically — the sector economics, the counterparties, the regulatory and political exposure, the revenue mechanisms (availability payment, usage or demand, regulated return) and the investor base that buys these assets at each stage of life.

By level. Foundation — distinguishes the main revenue mechanisms and what each implies for risk. Practitioner — analyses an asset within a sector they know: understands its demand drivers, regulatory regime and comparable transactions. Professional — assesses an asset in a sector or jurisdiction that is new to them by identifying what must be true for the revenue to arrive, and tests it. Leader — sets sector strategy or lending appetite, and owns the concentration risk that follows.

Evidence. A sector or asset analysis you produced, naming demand drivers and regulatory exposures; a revenue-mechanism comparison for a specific asset; a written view on a jurisdiction's regulatory risk that was tested by subsequent events.

The false positive. Sector familiarity mistaken for sector analysis. Having worked on eight roads does not mean a person can say what would make the ninth fail. The competency is analytical, not biographical.

2.3 Financial modelling

Definition. Building, auditing and using a model that represents a transaction faithfully enough to support a decision — structure, integrity, transparency of assumptions, and the discipline that lets someone else find and test any number in it. Depth in CMP-06 §6.

By level. Foundation — navigates a model, traces a calculation and updates inputs without breaking it. Practitioner — builds and maintains model sections to a defined standard: consistent formulae, no hard-coded values in calculation blocks, working checks, documented inputs. Professional — designs a model for a transaction: decides granularity, timing convention, circularity treatment and scenario architecture; audits others' models and finds the errors that matter; states

the model's limitations alongside its output. Leader — sets modelling standards, decides what model risk the organisation accepts, and requires independent review where reliance is material.

Evidence. A model you built, with its documentation, check sheet and assumption log; a model audit report you wrote, with findings and materiality; a scenario architecture you designed; a modelling standard in use in your organisation with your authorship.

The false positive. Speed and formula complexity. A rapidly built model with elaborate formulae and no checks is a liability, and the more impressive it looks the longer the error survives. The competency is visible in the checks, the assumption log and the auditability, which is exactly the part that takes the time.

2.4 Investment appraisal

Definition. Deciding whether a project is worth doing, using discounted cash flow and its companions, with the discount rate, the cash flow definition and the treatment of tax and inflation chosen deliberately and stated. Includes knowing which metric answers which question and where each one misleads.

By level. Foundation — computes net present value (NPV), internal rate of return (IRR) and payback correctly from a given cash flow. Practitioner — prepares an appraisal to procedure, applies the organisation's discount rate, and presents results with sensitivities. Professional — decides the appraisal basis: which cash flows are relevant, whose returns are being measured (project or equity), the discount rate and its justification, and the consistent treatment of inflation and tax. Identifies where IRR is a misleading answer — non-conventional cash flows, mutually exclusive projects of different scale or life — and says so. Leader — sets the appraisal standard and hurdle-rate policy, and owns the consequence of the incentives that policy creates.

Evidence. An appraisal you authored stating the cash flow definition, discount rate basis and inflation/tax convention; a sensitivity and breakeven analysis naming the variables the decision turns on; a documented case where you rejected a metric as unsuitable and explained what to use instead.

The false positive. Producing the metrics. The arithmetic is not the competency; the definition of what is being discounted, at what rate, and whose return it represents is the competency. An appraisal whose discount rate is "the one we always use" has not been appraised.

2.5 Capital structure

Definition. Deciding the shape of the funding — how much debt against equity, the tenor and amortisation profile relative to the asset's cash generation, the sequencing of drawdowns, and the consequences of the shape for resilience and for returns.

By level. Foundation — reads a sources-and-uses statement and a funding profile accurately. Practitioner — models alternative gearing and amortisation profiles and reports their effects on returns and cover ratios. Professional — recommends a structure for a specific transaction, stating the trade-off between equity return and the project's ability to survive a downside, and the point at which additional leverage stops being efficient. Leader — sets structuring policy, and decides the leverage the organisation will not exceed regardless of what a market will bear.

Evidence. A structuring paper comparing profiles with the reasoning for the recommendation; a downside resilience analysis at alternative gearing levels; a documented recommendation to reduce leverage against commercial pressure to increase it.

The false positive. Optimisation to a target return. Solving for the gearing that hits an equity return is an exercise in inversion. The competency is the judgement about what the structure does when revenue disappoints, which is the question the model was not asked.

2.6 Debt and equity

Definition. The instruments themselves: senior and subordinated debt, mezzanine, sponsor equity and shareholder loans, bonds and private placements, hedging arrangements — their terms, their ranking, their protections, and the behaviour of their holders when the project underperforms.

By level. Foundation — identifies instruments and their ranking in a capital stack. Practitioner — works with term sheets and facility documentation, tracks conditions and covenants, and models drawdown, interest and repayment mechanics correctly. Professional — negotiates or advises on terms with a view to how they will operate in a downside: which covenant bites first, what the cure rights are, and how the intercreditor arrangement behaves when parties disagree. Leader — sets the terms the organisation requires and concedes, and owns the portfolio consequence of that position.

Evidence. A term sheet you negotiated or reviewed with your annotations and rationale; a covenant analysis showing which test binds under stress; a waiver or restructuring you worked on, with the reasoning; hedging documentation and its modelled effect.

The false positive. Documentation familiarity. Knowing where a provision sits is administration. The competency is anticipating the instrument's behaviour under stress — and the honest test is whether the person has ever seen, or modelled, what happens when the covenant is breached.

— 3. Bankability, structures and the transaction

3.1 Project bankability

Definition. The composite judgement about whether lenders will fund a project on acceptable terms: revenue certainty, contract quality, counterparty strength, risk allocation, sponsor capability, and the resilience of cover under downside cases. Bankability is a conclusion assembled from the other competencies, not a measure in itself.

By level. Foundation — lists the factors lenders assess. Practitioner — assembles a bankability assessment against a checklist and identifies gaps in the documentation. Professional — forms and defends a view on whether a project is bankable, names the two or three issues that will decide it, and proposes the specific structural changes that would fix them. Leader — sets the institution's bankability standard and declines transactions that meet the checklist but fail the judgement.

Evidence. A bankability assessment you authored, with the issues ranked and the remedies proposed; a gap-closure plan that was implemented; a credit paper you wrote, and the outcome; a project you advised against, with the reasons recorded.

The false positive. Checklist completion. A complete checklist tells you the documents exist. The competency is knowing which two items on the list will actually determine the outcome, and being right often enough to be worth asking.

3.2 DSCR, LLCR and PLCR

Definition. The cover ratios by which lenders measure protection — the debt service cover ratio (DSCR), being cash flow available for debt service (CFADS) over scheduled debt service in the same period; the loan life cover ratio (LLCR), being the present value of CFADS over the loan's remaining life against debt outstanding; and the project life cover ratio (PLCR), the same construction over the project's full remaining life. The competency includes knowing what each ratio protects against and how each is manipulated.

By level. Foundation — calculates each ratio correctly and states its definition, including what is and is not in CFADS. Practitioner — models the ratios across a transaction's life, identifies the binding period, and reconciles ratio movements to their drivers. Professional — decides what the lock-up and default levels should be for a given risk profile, defines CFADS precisely in the documentation, and detects definitional games — items moved above or below the CFADS line, reserve accounts flattering a period, a discount rate choice inflating LLCR. Leader — sets minimum cover standards by asset class and owns the exceptions.

Evidence. A ratio analysis identifying the binding period and its driver; a negotiated CFADS definition and your rationale; a documented challenge to a ratio presentation you judged misleading; a downside case showing where cover is lost and what happens next.

The false positive. Calculation. The ratio is arithmetic once CFADS is defined; the competency is in the definition, the base case behind it and the downside at which it fails. A person who can produce the ratio but cannot say what would have to happen for it to breach has the arithmetic, not the competency.

3.3 PPPs and concessions

Definition. Structures in which a public authority contracts a private party to finance, build and operate an asset over a long term — the payment mechanism, the risk transfer intended and the risk transfer achieved, the performance and deduction regime, change and termination provisions, and handback.

By level. Foundation — describes the main PPP and concession models and how each is paid. Practitioner — works within a concession: models the payment mechanism including deductions, tracks performance obligations, and applies change provisions. Professional — assesses whether a proposed risk transfer is real — whether the private party can price and manage the risk it is being given — and identifies the provisions that will cause disputes in year eight. Advises on the termination and handback regime as a financeable proposition. Leader — sets policy on which concession structures the organisation will accept, and owns long-tenor political and regulatory exposure.

Evidence. A payment-mechanism model with the deduction regime implemented; an analysis of risk transfer versus risk pricing for a specific project; commentary on termination compensation and its effect on financeability; a lessons-learned note from an operating concession.

The false positive. Model familiarity. Knowing the taxonomy of PPP models is introductory material. The competency shows in the payment mechanism — the one document that decides whether the asset earns — and in whether the person has ever traced a deduction through to its cash effect.

3.4 Project contracts

Definition. The contract suite that makes the cash flows credible: the construction contract, the operations and maintenance agreement, the offtake or revenue agreement, supply agreements, and the interfaces between them. The competency is reading them as a financing package rather than as separate documents, and finding the gaps between them.

By level. Foundation — identifies the principal contracts and their commercial purpose. Practitioner — extracts and models the financial terms: liquidated damages, caps, indexation, payment terms, availability regimes. Professional — identifies mismatches across the suite — a liability cap below the loss it is meant to cover, an indexation basis that differs between revenue and cost, a force majeure definition that leaves relief on one side of a back-to-back — and negotiates them closed. Leader — sets the contractual conditions the organisation requires before it will lend or invest.

Evidence. A contract review memorandum identifying financial exposures with clause references; a back-to-back gap analysis; a negotiated change that closed a mismatch you found; a model faithfully implementing contractual mechanics.

The false positive. Legal review. Lawyers assess enforceability; this competency assesses financial consequence. The distinguishing question is whether the person can say what a clause is worth in cash and in which scenario it pays.

3.5 Financial close

Definition. Getting a transaction from agreed terms to funded: the conditions precedent, the diligence workstreams and their sequencing, document negotiation, the closing model and its agreement between parties, and the choreography of signing and first drawdown.

By level. Foundation — tracks conditions precedent and maintains the checklist accurately. Practitioner — runs a workstream to close: manages diligence inputs, resolves routine conditions, and maintains the closing model version control. Professional — runs or leads a close: sequences the critical path of the transaction, decides which conditions can be waived or deferred and on what terms, and holds the base-case model agreed across parties. Leader — decides whether a transaction is ready to close, and stops one that is not despite the cost of delay.

Evidence. A conditions-precedent tracker you owned through to satisfaction; a closing checklist and timetable you built; a documented decision to defer or waive a condition, with the mitigation; an agreed final base-case model with version control and sign-off.

The false positive. Presence at closings. Attending a completed transaction is exposure. The competency is in the critical path — knowing which condition would have delayed close and what was done about it two weeks earlier.

3.6 Commercial and financial risk

Definition. Identifying, allocating, mitigating and pricing the risks that determine whether the cash flows arrive: construction cost and delay, revenue and demand, counterparty credit, interest rate, inflation, currency, refinancing, regulatory and political risk. The competency is the connection between identifying a risk and doing something with it in the structure, the contracts or the price.

By level. Foundation — identifies the main risk categories and where each is usually allocated. Practitioner — runs risk analysis for a transaction: builds sensitivities and downside cases,

quantifies exposures, and tracks mitigation into the documentation. Professional — decides the risk case: which scenarios must be survived and which are tolerable, how correlated risks are treated together, and where the residual sits after all mitigation. Says what the project cannot survive. Leader — sets the institution's risk appetite for the asset class and owns the aggregate position.

Evidence. A downside and stress-case suite with the reasoning for each case; a risk allocation matrix with the mitigation and residual named; a decision to price or decline a specific risk; a case where your stress case was later approximated by events.

The false positive. Sensitivity tables. Flexing one variable at a time produces a grid, not a risk view; real downsides arrive correlated, and the single-variable table is the one presentation that guarantees they will not be seen. The competency is the combined case, and the willingness to name the scenario in which the project fails.

3.7 Sustainable finance

Definition. Financing structures and obligations tied to environmental and social outcomes: use-of-proceeds instruments and sustainability-linked pricing, lender E&S requirements and the diligence and monitoring they require, transition and physical climate risk in the cash flows, and disclosure obligations that vary by market.

By level. Foundation — describes the main instrument types and the difference between use-of-proceeds and performance-linked structures. Practitioner — implements the requirements: assembles E&S diligence inputs, models margin adjustments tied to performance indicators, and maintains reporting obligations. Professional — designs the sustainability provisions so that they are measurable and meaningful: selects indicators that are material and auditable, sets thresholds that require actual change, and resists structures whose targets would be met by ordinary course of business. Assesses E&S risk against recognised lender frameworks. Leader — sets the organisation's position, including declining transactions that are compliant and not credible.

Evidence. A sustainability-linked mechanism you specified, with indicator definitions and thresholds; an E&S diligence scope and its findings; a climate risk analysis integrated into cash flow rather than appended; a documented rejection of a proposed indicator as unmeasurable or immaterial.

The false positive. Labelling. An instrument with a sustainability label, a framework document and a second-party opinion may involve no change in behaviour whatsoever. The competency is in indicator design and materiality — whether meeting the target requires the borrower to do something they would not otherwise have done.

3.8 Islamic project finance

Definition. Structuring project financing consistent with Islamic finance principles — asset-backed or asset-based structures such as ijara (lease), istisna'a (construction and manufacture), murabaha (cost-plus sale), wakala (agency) and sukuk issuance — and running them alongside conventional tranches where a transaction has both. The competency includes the governance interface: what Shariah approval requires and how it constrains structuring.

By level. Foundation — describes the principal structures and how each generates a return without interest. Practitioner — works within an Islamic tranche: models its payment mechanics, tracks its documentation and maintains its distinctions from a conventional tranche. Professional — structures the Islamic element of a transaction, resolves intercreditor and ranking issues between Islamic and conventional tranches, and manages the approval process without

compromising the financing's commercial logic. Leader — sets the institution's approach and its relationships with Shariah governance bodies.

Evidence. An Islamic tranche you structured or modelled, with its mechanics documented; an intercreditor analysis across Islamic and conventional tranches; a structuring paper responding to Shariah board observations; a sukuk structure diagram you produced.

The false positive. Terminology substitution. Treating an Islamic structure as a conventional loan with different labels misses the asset ownership, risk-sharing and permissibility constraints that determine what may be done, and produces structures that fail at approval. The competency requires understanding why the structure is shaped as it is, not only that it is.

— 4. The data, digital and AI group

Five competencies, forming the same chain described in CMP-08: analysis produces a proposal, the workflow decides where it enters, governance constrains it, and a named human validates and owns it.

4.1 AI-enabled financial modelling

Definition. Using AI assistance within modelling work — generating or reviewing formulae, checking structure, drafting documentation, interrogating a model in natural language, testing for errors — with a clear position on what the model's integrity depends on and what may never be delegated.

By level. Foundation — uses approved assistance for defined tasks and checks output before use. Practitioner — incorporates assistance into modelling routines: error scanning, documentation drafting, consistency checking, with the results verified against the model itself. Professional — decides where assistance is admitted and where it is excluded — typically excluding the definition of the transaction logic, which is the part that has to be reasoned — and specifies the check that stands between assisted output and reliance. Leader — sets the standard for AI-assisted modelling, including disclosure and model-risk treatment, and owns the consequences.

Evidence. A documented workflow showing assisted steps and their checks; an error-detection routine and what it found; a documented decision on which modelling tasks are not delegated; a model with an assistance disclosure and validation record.

The false positive. Output volume. Producing model sections quickly with assistance and no verification step increases model risk with excellent presentation. The competency is the check design and the exclusions.

4.2 Predictive cash-flow analysis

Definition. Using data and quantitative technique to say something about future cash generation — demand or usage forecasting, cost and revenue trend analysis, probabilistic distributions of CFADS, early-warning indicators — and knowing when a data-driven forecast is inadmissible because the data does not carry the question.

By level. Foundation — interprets a probabilistic output correctly, including its confidence and its base. Practitioner — builds and runs the analysis, prepares data, tests fit, and reports central case with dispersion. Professional — chooses the technique for the question, back-tests against held-out periods, reports their own model's error and bias, and states where the analysis fails — typically

for assets without a comparable history, or where regulatory change breaks the series. Leader — sets what the organisation will forecast quantitatively and owns the data investment that makes it possible.

Evidence. An analysis with its data basis, validation and error characteristics stated; a back-test of your own forecasts; a documented refusal to produce a quantitative forecast where the data would not support it, and the alternative offered.

The false positive. Fitting the past. High explanatory power over history says nothing about predictive power, and a model with an impressive fit is the easiest thing in this discipline to sell to a committee. The competency shows in the out-of-sample test.

4.3 Digital due diligence

Definition. Using data and automated tooling to run diligence better — completeness checking across data rooms, contract term extraction and comparison, anomaly detection in historical financials, model audit assistance — and knowing which diligence conclusions may never rest on automated extraction alone.

By level. Foundation — uses the tooling as directed and verifies extracted items against source. Practitioner — runs digital diligence workstreams, reconciles extracted data to primary documents, and maintains an audit trail from finding to source. Professional — designs the diligence approach: what is automated, what sampling verifies the automation, and which findings require a human reading of the whole document because the meaning is in the interaction between clauses. Leader — sets diligence standards and owns the risk that automation created a false sense of completeness.

Evidence. A diligence plan distinguishing automated from manual scope with sampling rates; a reconciliation of extracted terms to source documents with the error rate found; a finding that automated extraction missed, and the process change that followed.

The false positive. Coverage claims. "All 400 contracts were reviewed" describes a process that ran, not a conclusion that holds. The competency is in the verification design — the sample that proves the extraction was right, and the honesty about what the sample can and cannot support.

4.4 Responsible AI

Definition. Governing AI use across the transaction lifecycle: what data may be entered into which tool, particularly counterparty and confidential deal information; where AI-derived analysis must be disclosed to a credit committee or investor; how bias is considered where models touch counterparty assessment; and which uses are prohibited.

By level. Foundation — follows the policy, especially on confidential data. Practitioner — applies it in live transactions, discloses assistance where required, and escalates outputs that cannot be explained. Professional — sets the position for a transaction or team: data handling, disclosure convention, explainability as a precondition of reliance, and named prohibitions. Leader — owns the governance framework, its procurement criteria and its incident handling, and prohibits uses the organisation would find profitable.

Evidence. A transaction-level AI use position you authored; a disclosure convention adopted in credit or investment papers; a documented prohibition and its rationale; an incident review with the resulting control change.

The false positive. Policy possession. A policy nobody has tested against a real decision is a document. The evidence is a decision that cost something.

4.5 Human validation

Definition. Checking AI-derived or automated output before it is relied upon, and owning it as your own work — designing a check capable of catching the error that matters, performing it, recording it, and being answerable for what you signed. In a financing context the outputs at issue are model results, extracted contract terms, forecast cash flows and drafted analysis.

By level. Foundation — performs the prescribed check and escalates failures. Practitioner — validates against independent sources within their scope and records what was checked and found. Professional — designs validation per output type: the independent source, the sample, the tolerance, and the errors the check would not catch. Rejects output that cannot be validated in the time available. Leader — sets the validation standard so it cannot be discharged by signature, and owns validated failures as failures of the standard.

Evidence. A validation protocol naming source and tolerance per output type; signed validation records including findings that stopped something; a rejection of unvalidatable output and its replacement; a revised check following a miss.

The false positive. The confirmation tick. The house position is that AI proposes and the professional disposes — which is only true where the disposal is an independent check by someone who could have caught the error and would have said so.

— 5. Where the boundaries actually lie

Project finance and infrastructure finance. The first is a financing technique and a risk-allocation discipline; the second applies it to a specific asset class with its own economics and counterparties.

Capital structure and debt and equity. Structure is the shape — how much, over what tenor, in what sequence. Debt and equity are the instruments that implement the shape and behave in particular ways when things go wrong.

Investment appraisal and financial modelling. Appraisal is the decision and its basis; the model is the instrument. A flawless model can implement an incoherent appraisal basis, and usually does so invisibly.

Bankability and the cover ratios. The ratios are measures within the judgement. A project can present strong ratios in a base case that no lender believes, and the competency is knowing that the base case is the question.

PPPs and project contracts. A PPP or concession is an ownership and payment structure; project contracts are the suite implementing it. The payment mechanism sits at the join and is where most value is won or lost.

— 6. How this goes wrong

All nineteen are claimed at leader level. No individual has that spread; Islamic project finance and sustainable finance in particular are specialisms that many strong practitioners have never

touched. Say so. An honest foundation-level statement costs nothing and buys credibility for the claims that matter.

Modelling evidence is a model file with no reasoning. A workbook demonstrates that a workbook exists. Evidence is the check sheet, the assumption log, the audit findings and the version history.

Cover-ratio competence is evidenced by a formula. The formula is public. Evidence is a negotiated CFADS definition, a binding-period analysis, or a challenge to a presentation that flattered a period.

Transaction experience substitutes for competence. "Worked on twelve closings" is a duration claim. Which condition would have delayed close, and what did you do about it before it did?

The false positive is the candidate's whole portfolio. Checklist bankability assessments, sensitivity tables in place of stress cases, extraction coverage in place of verification. Each of these is real work, performed at practitioner level, presented as professional-level judgement — and an assessor will place it correctly, so the candidate loses the credit they would have received for claiming it accurately.

Islamic finance structures are described as conventional finance with different names. This is the single most common error in the set, and it is visible immediately to anyone who works in the field.

Sustainability provisions are evidenced by the presence of a label. Name the indicator, its definition, its threshold and what the borrower had to change.

— 7. Checklist

For a candidate assembling evidence, or an assessor reviewing it:

- [] Have all nineteen competencies been addressed, including honest foundation-level statements?
- [] Does each have a **dated artefact** with an identifiable personal contribution?
- [] For modelling: is there a check sheet, an assumption log and an audit trail — not only a model?
- [] For appraisal: is the cash flow definition, discount rate basis and inflation/tax convention stated?
- [] For cover ratios: is there a CFADS definition, a binding period, and a downside at which cover fails?
- [] For bankability: are the two or three deciding issues named, or only a checklist?
- [] For risk: are there combined stress cases, not single-variable sensitivities?
- [] For the AI competencies: is the evidence a **decision** — where output enters, what check stands between it and reliance, what was excluded?
- [] Is human validation evidenced by a protocol and findings rather than a signature?
- [] Is every artefact dated, with anything older than three years flagged as historical?

— Related

- [CMP-01 – The PCI competency framework – overview](#) — what the set is for and how each entry is structured
- [CMP-02 – Competency levels and how they are evidenced](#) — the four-level scale used throughout
- [CMP-06 – Cost and commercial competencies in depth](#) — the commercial cluster shared across credentials
- [CMP-08 – Data, digital and AI competencies in depth](#) — the five-competency chain from analysis to validation
- [CMP-10 – Assessing competence – evidence, rubrics and moderation](#) — how these claims are tested
- [CER-02 – The credential suite: PCL-AI, PFL-AI, PML-AI](#) — how the three credentials differ and how to choose

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the PFL-AI seeded competency set, nineteen items, reproduced here in the published order and wording.
- [docs/books/pfl-ai/COMPETENCY_MAP.md](#) — the PFL-AI competency map, which records that all nineteen published competencies are covered and that no examination blueprint exists.
- Lender environmental and social frameworks in common use in project finance, referenced generically in §3.7 and described in our own words; no framework text is reproduced and no endorsement is implied.

— Status and version

Founding-stage document · Version 1.0 — effective date to be confirmed · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

CMP-05 · COMPETENCY FRAMEWORKS

PML-AI: the twenty-four competencies

Each competency of the PCI Project Management Leader – AI — defined, levelled, evidenced, and separated from its look-alike

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

The PCI Project Management Leader – AI credential publishes twenty-four competencies, the widest of the three sets. This document takes each one in turn and states what it is in terms of decisions, what it looks like at each of the four levels, the named artefacts that evidence it, and the common false positive — the thing that is routinely mistaken for the competency and is not it.

— About this document

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Examples. All worked examples are illustrative and fictional. Figures are chosen to demonstrate a method, not to describe any real project, organisation or market. Sample examination items, where they appear, are study material maintained separately from any live examination bank.

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PML-AI: the twenty-four competencies

from its look-alike.

In one paragraph. The PCI Project Management Leader – AI credential publishes twenty-four competencies, the widest of the three PCI sets. This document takes each one in turn and states what it is in terms of decisions, what it looks like at each of the four levels, the named artefacts that evidence it, and the common false positive — the thing routinely mistaken for the competency and which is not it. It is the working reference for anyone assembling evidence, specifying a delivery role, or assessing a claim against the PML-AI standard.

Who this is for. Candidates for PML-AI; delivery directors, programme managers and heads of project management office designing roles and development plans; assessors applying the standard to evidence they did not collect.

— 1. How to read this document

The twenty-four competencies below are the published PML-AI set, in the published order, unaltered. Each entry follows the structure in CMP-01 §3: **definition**, **by level** (the scale is defined in CMP-02 §2 and not restated), **evidence**, and **the false positive**.

Two notes on the text itself. First, **no examination blueprint has been published for PML-AI** — no domain weightings, no item counts [[CONFIRM: PML-AI examination blueprint — weightings and item counts](#)]. This set is the authoritative statement of what the credential claims; it is not a weighted syllabus. Second, the platform stores competency 16 as Benefits realization; this framework is written in British English and renders it benefits realisation. Same competency, spelling only.

Breadth is the defining feature of this set: it runs from initiation to closure and across three delivery approaches. Breadth is not difficulty — see CMP-01 §2. It does mean that no realistic candidate is at leader level across twenty-four competencies, and a portfolio claiming otherwise reads as a portfolio that has not been thought about.

Acronyms are expanded on first use: project management office (PMO), work breakdown structure (WBS), critical path method (CPM), request for proposal (RFP), key performance indicator (KPI), estimate at completion (EAC).

— 2. Leading and framing the project

2.1 Project management

Definition. Running a project as an integrated system rather than a set of parallel activities: holding scope, schedule, cost, risk, quality and resource in a single coherent position, and making the trade-off explicit when one has to move.

By level. Foundation — executes assigned elements to plan and reports status accurately. Practitioner — manages a project or workstream end to end within an established method, and keeps the integrated position current. Professional — manages a project whose method must be chosen rather than inherited, and makes trade-offs visible to decision-makers as options with consequences rather than as requests for more time. Leader — sets how projects are run across an organisation, and owns the outcome when the method proves inadequate.

Evidence. A project management plan you authored with the integration between elements visible; a trade-off paper presenting options with consequences and the decision taken; a recovery plan for a project you inherited in difficulty, with the diagnosis.

The false positive. Coordination. Running meetings, chasing actions and consolidating other people's status is administration performed capably. The competency shows at the trade-off: when scope, cost and date cannot all hold, does this person present the choice clearly enough for someone to make it?

2.2 Project leadership

Definition. Getting a group of people who do not report to you to do difficult work in a specified direction — setting a direction people can act on, making decisions with incomplete information, absorbing pressure without transmitting panic, and being the person who says the unwelcome thing first. Depth in CMP-09.

By level. Foundation — works effectively within a team and raises issues candidly. Practitioner — leads a small team through routine delivery; communicates direction and makes decisions inside their scope. Professional — leads a project team through genuine difficulty: names the problem before it is obvious, holds an unpopular position where the evidence supports it, and changes position publicly when it does not. Leader — builds delivery leadership in others, sets the culture in which bad news travels quickly, and is accountable for what the team could not say.

Evidence. A decision made under uncertainty with the reasoning recorded at the time; an escalation you initiated that was unwelcome; a team you developed, with what changed; a project you stopped or recommended stopping.

The false positive. Seniority and span of control. Managing thirty people is a resourcing fact. Leadership evidence involves a moment where the leader carried a cost — reputational, political or personal — for a position that turned out to be right, or wrong and honestly owned.

2.3 Project initiation

Definition. Establishing a project properly before work starts: the objective, the scope boundary, the sponsor and governance, the success criteria, the assumptions and constraints, the initial funding and the authority to proceed — captured in a charter someone has actually signed.

By level. Foundation — prepares initiation documentation from provided inputs. Practitioner — runs initiation for a project: assembles the charter, identifies stakeholders, establishes the governance and secures approval. Professional — challenges the premise at initiation: tests whether the stated objective is the real objective, whether success criteria are measurable, and whether the sponsor has the authority the project will need. Refuses to start on an unresolvable ambiguity. Leader — sets initiation standards, and stops projects at the gate rather than after twelve months of spending.

Evidence. A charter you authored, with measurable success criteria and named assumptions; a documented challenge at initiation that changed the scope or the sponsor; an initiation gate you failed a project at, with reasons.

The false positive. Document production. A complete charter pack assembled from a template records that the process ran. The competency is the challenge: what did you ask at initiation that changed something?

2.4 Business cases

Definition. Building or testing the case for investment: the problem, the options considered, the costs and benefits with their bases, the risks, the assumptions the case depends on, and the point at which the case stops being true.

By level. Foundation — compiles inputs and checks internal consistency. Practitioner — prepares a business case to the organisation's standard, including option comparison and financial analysis. Professional — challenges the numbers in a case, including their own: tests whether the benefit is attributable to the project, whether the counterfactual is honest, whether the cost base is complete, and states the conditions under which the case fails. Leader — sets the standard for cases, and requires post-implementation comparison of claimed to realised benefit, which is where optimism gets priced.

Evidence. A case you authored with option comparison, benefit attribution and stated break conditions; a challenge you made to another party's case, with the change that followed; a post-implementation review comparing claimed to realised benefits.

The false positive. Assembling the case document. The competency is in benefit attribution — whether the benefit would have occurred anyway, whether it is double-counted elsewhere, and whether anybody has agreed to be accountable for delivering it.

2.5 Governance

Definition. Deciding who decides: the decision rights, the thresholds at which authority changes hands, the forums and their real remits, the reporting that feeds them, and the assurance that tests whether the reporting is true.

By level. Foundation — operates within the governance framework and prepares papers to standard. Practitioner — runs a governance cycle: schedules gates, prepares decision papers, tracks decisions and follows through on actions. Professional — designs governance for a project: right-sizes the forums, sets thresholds that reflect real risk, ensures decisions are made by people who can carry them, and removes bodies that consume effort without deciding anything. Leader — governs the governance across a portfolio, including the willingness to dissolve a committee and the discipline of assurance that is genuinely independent.

Evidence. A governance framework you designed, with decision rights and thresholds; decision records where the forum actually chose between options; an assurance review you commissioned or acted on; a governance body you disbanded and why.

The false positive. Chairing meetings. A well-attended steering committee that receives updates and approves what has already happened is a ceremony. The test is whether the governance has changed a decision in the last six months — and whether anyone can name it.

2.6 Scope and requirements

Definition. Establishing what the project will and will not deliver, in terms specific enough to be verified: requirements elicitation and specification, traceability from need to deliverable, the scope boundary and its exclusions, and the discipline that keeps additions visible.

By level. Foundation — documents requirements accurately and maintains traceability records. Practitioner — elicits and specifies requirements for a scope, builds a WBS that decomposes the work completely, and manages the requirements baseline. Professional — resolves conflicting requirements between stakeholders, decides what is out of scope and defends the exclusion, and specifies acceptance criteria that will not be arguable at handover. Leader — sets requirements and scope standards, and carries the consequence when a scope boundary proves wrong.

Evidence. A requirements specification with acceptance criteria and traceability; a WBS and dictionary you authored; a documented exclusion that held under pressure; a requirements conflict you resolved, with the decision basis.

The false positive. Requirements volume. Four hundred requirements, untraced and untestable, is a document. The competency shows in acceptance criteria — whether it will be possible to say objectively, at handover, that a requirement was met.

— 3. Delivering the work

3.1 Planning and execution

Definition. Turning scope into a credible plan and then running it: sequence and dependency, resource and capacity, milestone architecture, the update cycle, and the response when the plan and reality diverge. Depth in CMP-07.

By level. Foundation — plans and executes assigned work, and reports variance honestly. Practitioner — builds and runs the plan for a project or workstream, maintains it through change, and explains variance in terms of cause. Professional — designs the planning approach for the work in hand, decides the level of planning detail that is useful rather than impressive, and replans deliberately when the plan stops being a description of what will happen. Leader — sets planning standards, and enforces the distinction between a plan and a target.

Evidence. A plan you built with its basis documented; a replan with the trigger, the options and the decision; a milestone architecture that survived contact with delivery; a variance analysis whose diagnosis was later confirmed.

The false positive. Plan detail. A twelve-thousand-line schedule maintained fortnightly may be an expensive record of the past. The test is whether the plan is used to make decisions, and whether anyone can say what it currently predicts and why.

3.2 Stakeholder leadership

Definition. Identifying who can affect or is affected by the project, understanding what each actually needs as opposed to what they ask for, and managing the relationships so that decisions get made and commitments hold. Depth in CMP-09.

By level. Foundation — communicates clearly and maintains stakeholder records. Practitioner — runs stakeholder engagement for a scope: maps interests and influence, tailors communication,

and manages routine expectation. Professional — handles conflict between powerful stakeholders with incompatible objectives: surfaces the conflict rather than absorbing it, and brings it to a forum that can resolve it. Says no to a stakeholder who matters, and keeps the relationship. Leader — owns the organisation's most difficult delivery relationships and builds the capability in others.

Evidence. A stakeholder analysis with influence and interest, and what you did differently as a result; a conflict you surfaced and how it was resolved; a documented refusal to a senior stakeholder and the subsequent relationship; a communication regime you designed and the behaviour it changed.

The false positive. The stakeholder register. A matrix of names, categories and colour codes is an artefact that took an afternoon. The competency is visible in a conversation nobody wanted to have.

3.3 Procurement

Definition. Deciding what to buy and from whom: make-or-buy and packaging strategy, market engagement, specification and tender documentation, evaluation criteria and their application, and the transition from award into delivery.

By level. Foundation — supports procurement processes and maintains documentation. Practitioner — runs a procurement to process: prepares the RFP, manages the tender period, applies the evaluation model and documents the award. Professional — decides packaging strategy — how work is split into contracts and where the interface risk consequently sits — and designs evaluation criteria that select for what the project needs rather than for the best proposal writer. Leader — sets procurement strategy across a portfolio, owning the supply-market consequences of it.

Evidence. A packaging strategy with the interface implications analysed; an evaluation model you designed, with weighting rationale; a tender you ran end to end with the award recommendation; a post-award review of whether the evaluation predicted performance.

The false positive. Process compliance. Running a compliant tender is necessary and is not the competency. The competency is packaging and evaluation design — the two decisions taken before the process starts that determine whether the right supplier can win.

3.4 Contracts and commercial delivery

Definition. Managing the contract after award so the commercial position stays defensible: obligations and notices, change and variation, payment and valuation, performance regimes, and the record that substantiates entitlement. Depth in CMP-06.

By level. Foundation — administers contract processes and maintains records. Practitioner — manages contract delivery for a scope: issues notices in time, values changes, maintains the commercial register and applies the performance regime. Professional — decides the commercial position on contested matters: what to concede, what to reserve, what to disclose as exposure; assembles the contemporaneous record; and manages the interface with legal and commercial functions without outsourcing the judgement. Leader — sets commercial delivery standards and owns aggregate exposure.

Evidence. A change control log with the trail from instruction to agreed value; a notice issued within a time-bar and the entitlement it preserved; an exposure report separating forecast from unquantified exposure; a negotiated settlement with the basis recorded.

The false positive. Contract knowledge without record discipline. Being able to cite the mechanism does not substantiate a claim eighteen months later. The competency is whether the daily records, the delay evidence and the correspondence exist and are organised.

3.5 Cost, schedule and risk integration

Definition. Holding cost, schedule and risk as one position rather than three reports: a schedule that reconciles to the cost forecast, risk exposure that reconciles to contingency, and a change that moves all three consistently.

By level. Foundation — produces the three sets of data accurately. Practitioner — reconciles them for a scope and identifies inconsistencies between forecasts. Professional — designs the integration: common breakdown structures, a single change process affecting all three, contingency draw-down tied to risk retirement, and an EAC that reflects the schedule position rather than contradicting it. Leader — requires integration across a portfolio and refuses reporting that presents the three positions as independent.

Evidence. A reconciliation between schedule forecast and cost forecast, with the differences explained; a contingency drawdown analysis tied to risk retirement; an integrated change control process you designed; a case where you identified that the three positions could not all be true.

The false positive. Three reports in one pack. Physical adjacency is not integration. The diagnostic question: if the schedule slips two months, does the cost forecast move — automatically, traceably, and by an amount someone can explain?

3.6 Predictive delivery

Definition. Delivering under a plan-driven approach: a defined scope baselined up front, CPM scheduling, baseline change control, and progress measured against the plan. The competency includes knowing the conditions under which the approach works — stable requirements, understood technology — and recognising when those conditions are absent.

By level. Foundation — works within a baselined plan and reports progress against it. Practitioner — runs predictive delivery for a scope: maintains the baseline, controls change, and measures progress with defined rules of credit. Professional — decides when predictive delivery is the right approach and defends the baseline against uncontrolled change, while recognising when the baseline has become fiction and requiring a rebaseline rather than managing to a plan nobody believes. Leader — sets the standard for baseline integrity and decides when a portfolio's approach must change.

Evidence. A baseline you established with its change control regime; a rebaseline you recommended, with the trigger and the analysis; a progress measurement regime with rules of credit; a change you refused and why.

The false positive. Defending an obsolete baseline. Rigour and rigidity look identical from outside and are opposite competencies. The distinguishing test is whether the person has ever concluded that their own baseline no longer represented the work, and said so.

3.7 Agile delivery

Definition. Delivering iteratively against an ordered backlog with working increments, empirical progress measurement and short feedback cycles — and knowing what the approach requires to

function: an available decision-maker, a genuinely ordered backlog, a real definition of done, and the organisational willingness to change scope rather than date.

By level. Foundation — works within an agile team and understands the events and artefacts. Practitioner — runs or facilitates delivery for a team: maintains the backlog, protects the increment, measures throughput, and surfaces impediments. Professional — establishes agile delivery where it does not yet function: secures a decision-maker with real authority, insists on a definition of done that includes the parts nobody enjoys, and reports empirical forecast ranges rather than commitments the data does not support. Leader — decides where the organisation delivers this way and changes the surrounding governance and funding to make it possible.

Evidence. Throughput or cycle-time data with the forecast you derived and the outturn; a definition of done you established and enforced; an impediment escalated to and resolved at organisational level; a backlog ordering decision that cost a stakeholder something.

The false positive. Ceremonies. Running stand-ups, retrospectives and a board is a process performed. The competency test: can the team release, does the definition of done include testing and documentation, and does anyone act on the empirical data — or is the date still fixed and the scope still fixed?

3.8 Hybrid delivery

Definition. Combining predictive and adaptive approaches in one delivery where the work genuinely differs — typically a fixed-date, fixed-interface infrastructure element alongside iteratively developed software or process — and designing the joints: integration points, a common reporting position, and one governance forum that can read both.

By level. Foundation — works within a hybrid arrangement and understands both cadences. Practitioner — operates across both: maintains a plan for the predictive elements and a backlog for the adaptive ones, and manages the dependencies between them. Professional — designs the hybrid: decides which parts are which, defines the interface contract between them, and creates a single integrated reporting position rather than two incompatible ones. Leader — sets the organisation's approach and resolves the funding and governance mismatch that hybrids always expose.

Evidence. A delivery approach decision with the reasoning per element; an interface or dependency management design between cadences; an integrated report both audiences accepted; a governance forum that worked across both.

The false positive. Method mixing without design. "We're agile within a stage gate" often means neither approach functioning: no baseline integrity and no empirical measurement. Hybrid competence is visible in the joint, and the joint is a designed artefact.

3.9 Quality and change management

Definition. Two connected disciplines: assuring that what is delivered meets the specification (quality planning, assurance, control, acceptance and defect management), and controlling changes to what was agreed (change assessment, authority, and traceability into scope, cost, schedule and risk).

By level. Foundation — applies quality procedures and records changes accurately. Practitioner — runs quality control and change control for a scope: inspection and test regimes, defect tracking, change assessment and approval routing. Professional — designs both for a project: what is assured versus inspected, acceptance criteria that will hold at handover, and a change process that

makes the cumulative effect of small changes visible rather than letting it accumulate invisibly. Leader — sets quality and change standards, and owns the cost of quality as a managed number.

Evidence. A quality plan distinguishing assurance from control; a defect trend with the intervention and its effect; a change register showing cumulative impact against the original baseline; a rejected change with its assessment.

The false positive. Change log volume. A complete register that nobody aggregates hides the number that matters — the cumulative movement from the original position. The competency shows in the aggregate view, and in whether anyone has ever been surprised by it.

— 4. Value, closure and the AI group

4.1 Benefits realisation

Definition. Ensuring the project produces the outcome it was funded for: defining benefits so they can be measured, assigning ownership to people who can influence them, establishing the baseline before the change, and measuring after handover — including when the answer is unwelcome.

By level. Foundation — collects benefit measurement data as defined. Practitioner — maintains the benefits profile, tracks measures and reports realisation. Professional — defines benefits so that realisation is falsifiable: metric, baseline, measurement method, owner and date; separates project outputs from business outcomes; and reports shortfall honestly. Leader — holds the organisation to post-implementation measurement, and feeds the result back into how future cases are appraised.

Evidence. A benefits profile with baselines captured before the change; a post-implementation review comparing claimed to realised, including a shortfall; a benefit you reclassified as unattributable; a change to business case standards driven by realisation data.

The false positive. Benefit declaration at closure. Stating in the closure report that benefits will be realised, and then dissolving the team, is the standard failure across the discipline. The competency requires a measurement that happened after the project ended, and a named person who owned the number.

4.2 Sustainability

Definition. Managing the environmental and social dimensions of delivery as a delivery matter: whole-life impact in design and option choice, emissions and resource use during execution, ethical and responsible supply chain, community and workforce effects, and the legacy or handback condition of the asset.

By level. Foundation — complies with requirements and records data accurately. Practitioner — implements the plan: monitors and reports indicators, applies supply-chain requirements, and manages community obligations. Professional — integrates sustainability into decisions rather than reporting: brings whole-life impact into option selection, sets supply-chain requirements that change what is bought, and states the trade-off with cost and programme explicitly instead of asserting there isn't one. Leader — sets the organisation's delivery standard and accepts the cost consequences.

Evidence. An option analysis where an impact assessment changed the selection; supply-chain requirements you specified and their effect on procurement; a measurement regime with data and outcomes; a handback or legacy plan you authored.

The false positive. Reporting against indicators. Collecting and publishing data is measurement. The competency is whether a decision came out differently — and if none did, the sustainability work was reporting.

4.3 Project closure

Definition. Ending a project properly: completing and accepting deliverables, closing contracts and resolving final accounts, transferring the asset and its operating knowledge, releasing resources, archiving records so they are retrievable, and capturing lessons in a form somebody will use.

By level. Foundation — completes closure tasks and documentation. Practitioner — runs closure for a scope: acceptance, contract closeout, handover documentation and archive. Professional — manages contested closure: disputed final accounts, incomplete acceptance, an operator unwilling to receive the asset — and closes it rather than letting it drift into a maintenance backlog. Ensures lessons are specific enough to act on. Leader — sets closure standards and prevents the organisational habit of leaving projects ninety-five per cent closed indefinitely.

Evidence. A closure report with acceptance status, final account position and outstanding items; a disputed final account you settled, with the basis; a handover and training package the operator accepted; a lesson that changed a subsequent project, traceably.

The false positive. The closure report. A completed template records that a process ran. The competency is visible in the hard closures — the contract with an unresolved claim, the operator who did not want the asset — and in whether any lesson was ever adopted.

4.4 AI-enabled project management

Definition. Designing how delivery management uses artificial intelligence: which tasks are delegated (status synthesis, risk identification prompts, document drafting, schedule analysis, meeting capture), what the model receives, where its output enters the management process, and what check stands between that output and a decision. Depth in CMP-08.

By level. Foundation — uses approved tools within the defined process and checks output before use. Practitioner — operates AI-assisted workflows end to end and recognises implausible output. Professional — designs the workflow: task selection, input and output specification, the check, and the tasks that are excluded — usually the ones involving a judgement someone must own. Leader — sets the delivery function's AI operating model, admits tools, and decides where the organisation will not use AI.

Evidence. A process map showing assisted steps with check points; a tool evaluation with criteria and decision; a task removed from an assisted workflow and why; an incident review after AI output reached a decision.

The false positive. Tool adoption. The competency is the design — where output enters, and what would catch it if it were confidently wrong.

4.5 Decision intelligence

Definition. Improving how decisions are actually made: framing the decision and its options, defining what evidence would change the answer, using analysis proportionate to the stakes, guarding against the predictable biases of a delivery environment, and recording the decision with its basis so it can be reviewed later.

By level. Foundation — prepares decision inputs accurately and completely. Practitioner — structures decisions for a forum: options, criteria, evidence, recommendation, and a decision record. Professional — designs the decision process for consequential choices: identifies the decision-maker, frames genuine alternatives rather than one option and two straw men, names what would falsify the recommendation, and applies analysis proportionate to the stakes. Leader — sets decision standards across a portfolio, and requires periodic review of past decisions against outcomes.

Evidence. A decision paper with genuine alternatives and falsification conditions; a decision record with basis and assumptions, later reviewed against outcome; a decision you reopened when an assumption failed; a bias you designed against and the mechanism you used.

The false positive. Analytical volume. A hundred-page options appraisal supporting a foregone conclusion is elaborate ratification. The competency test: did the analysis have a realistic chance of producing a different answer, and has this person ever changed their recommendation because of it?

4.6 Digital delivery

Definition. Running delivery on a coherent digital toolchain: which system holds each data item, how they connect, how status flows from source to report without manual re-keying, and how the resulting record is retained and retrievable.

By level. Foundation — uses the toolchain correctly and maintains data quality in their scope. Practitioner — administers delivery tooling: configures workflows, maintains integrations, resolves discrepancies between systems. Professional — designs the delivery information architecture: source of truth per item, integration points, definitions agreed across functions, and the deliberate removal of duplicate systems. Leader — sets the standard across the portfolio, including data retention, and resolves the political problem that every function prefers its own system.

Evidence. An information architecture or data-flow map you produced; an integration that removed manual re-keying, with the error rate before and after; a duplicate system you retired; a data definition dispute you resolved.

The false positive. Tool count. A rich estate of poorly connected tools produces more reconciliation work than the manual process it replaced. The competency is in the architecture — one source per item, and lineage that survives an audit.

4.7 Automation

Definition. Removing manual steps from the delivery management cycle without removing the control: deciding what to automate, designing so failures are loud rather than silent, and maintaining it as surrounding systems change. Shared in substance with the PCL-AI competency of the same name; see CMP-08 §5.

By level. Foundation — runs automated routines and reports failures. Practitioner — builds and maintains automation for a defined task with error handling and a reconciliation that proves the

output. Professional — decides what should and should not be automated on frequency, stability and the cost of a silent error; builds break-detection; documents it so it survives the author. Leader — sets the automation standard including ownership, change control and the prohibition on single-person dependencies.

Evidence. An automated process with documentation, error handling and reconciliation; a break you designed that later caught a real failure; a decision not to automate, with reasoning; a handover in which someone else took the automation over.

The false positive. Script count. Automation that fails silently and produces a plausible wrong number increases risk while appearing to increase efficiency. When this breaks, how will anyone know?

4.8 Responsible AI

Definition. Governing AI use in delivery: what project, personal and commercial data may enter which tool; where AI-derived material must be disclosed to a board, a client or a supplier; how bias is considered where models touch people — resourcing, performance, supplier evaluation; and which uses are prohibited.

By level. Foundation — follows policy, particularly on confidential and personal data. Practitioner — applies it in live delivery, discloses assistance where required, and escalates unexplainable output. Professional — sets the project position: permitted uses, data handling, disclosure convention, and explainability as a condition of reliance rather than a preference. Leader — owns the governance framework, procurement criteria and incident handling, and prohibits uses the organisation would find convenient.

Evidence. A project-level AI use position you authored; a disclosure convention adopted in reporting; a documented prohibition with rationale; an incident review with the control change that followed.

The false positive. Policy possession. The evidence is a decision that cost something — a tool not adopted, a use case declined, a disclosure that invited a difficult question. AIG-08 and AIG-09 own the underlying framework.

4.9 Human accountability

Definition. The named person who answers for a decision or a deliverable, including one produced with AI assistance or by an automated process, and including work they did not personally perform. Broader than the PCL-AI and PFL-AI competency human validation: validation is the act of checking; accountability is the standing answerability for the outcome and the design of a system in which someone is always answerable.

By level. Foundation — takes responsibility for their own work, including its errors, and does not attribute an output to a tool. Practitioner — owns delegated and assisted outputs within their scope: checks what must be checked, and answers for what they release. Professional — designs accountability into a workflow so no output reaches a decision without a named owner; refuses to release output they cannot stand behind; and does not deflect to "the system said". Leader — ensures the delivery organisation has no accountability gaps — no decision taken by a process with no owner — and answers for failures of that design personally.

Evidence. An accountability map for a workflow, with named owners per output; a decision you owned whose analysis was produced by a model or a team, including how you satisfied yourself; a

case where you declined to release output you could not stand behind; a post-incident review in which you owned a failure in a process you designed.

The false positive. Sign-off. A signature on a document produced by someone or something else, without a statement of what was checked, is a record of a signature. The house position is that AI proposes and the professional disposes: disposal means a person is answerable, by name, when it is wrong.

— 5. Where the boundaries actually lie

Project management and project leadership. The first is the integrated system of scope, cost, schedule and risk; the second is what makes people act. Strong evidence for one is routinely offered for the other, and it does not transfer.

Governance and project management. Governance decides who decides; project management operates within those decision rights. A person who has designed governance has done something different from a person who has followed it well.

Predictive, agile and hybrid delivery. Three distinct competencies, not one spectrum. The most common gap is a candidate strong in one, describing the other two in vocabulary rather than in decisions.

Business cases and benefits realisation. The case forecasts benefit; realisation measures it after handover. Almost every portfolio has evidence of the first; the second is where candidates separate.

Human accountability and human validation. Validation is the act of checking (see CMP-03 §3.6 and CMP-04 §4.5). Accountability is answering for the outcome, including for checks performed by others.

— 6. How this goes wrong

Twenty-four competencies, twenty-four leader-level claims. Immediately implausible, and it devalues the claims that were true. Breadth is the property of the set, not of the individual.

One project supplies all the evidence. A single large project can legitimately evidence many competencies, but if every artefact carries the same project name and the same two years, the portfolio demonstrates a role rather than a repertoire. Show a second context, especially for the delivery-approach competencies.

Agile competence is evidenced by ceremonies and certificates. Evidence is throughput data with a forecast and an outturn, a definition of done that was enforced, and a decision that changed scope rather than date.

Predictive competence is evidenced by baseline defence alone. Defending a baseline is half the competency; recognising when the baseline has become fiction is the other half, and the harder half.

Hybrid delivery is claimed because both approaches are present in the organisation. Hybrid competence is the designed joint between them. If nobody designed the joint, there is no hybrid competence in the building.

Benefits realisation is evidenced by the business case. The case is a forecast. Realisation evidence happens after handover and frequently reports a shortfall. A portfolio in which every benefit was realised in full is not a strong portfolio; it is an unmeasured one.

Human accountability is evidenced by job title. Accountability is demonstrated at the moment something went wrong. If nothing in the portfolio ever went wrong, the portfolio is incomplete, not exemplary.

— 7. Checklist

For a candidate assembling evidence, or an assessor reviewing it:

- [] Have all twenty-four competencies been addressed, with honest levels including foundation?
 - [] Does the evidence span more than one project and more than one delivery approach?
 - [] For predictive: is there both baseline defence **and** a rebaseline judgement?
 - [] For agile: is there empirical data and an enforced definition of done, rather than ceremonies?
 - [] For hybrid: is there a designed interface between cadences, and one integrated report?
 - [] For governance: can you name a decision the governance changed?
 - [] For benefits: is there a measurement taken **after** handover, including a shortfall?
 - [] For business cases: is benefit attribution addressed, not just benefit totals?
 - [] For the AI competencies: is the evidence a decision — where output enters, what check, what excluded?
 - [] For human accountability: is there an instance of owning a failure, by name?
 - [] Is every artefact dated, with anything older than three years flagged as historical?
-

— Related

- [CMP-01 – The PCI competency framework – overview](#) — what the set is for and how each entry is structured
- [CMP-02 – Competency levels and how they are evidenced](#) — the four-level scale used throughout
- [CMP-07 – Planning, scheduling and risk competencies in depth](#) — planning, integration and risk at working depth
- [CMP-08 – Data, digital and AI competencies in depth](#) — the AI chain from analysis to accountability
- [CMP-09 – Behavioural and leadership competencies](#) — leadership, stakeholder and accountability behaviours
- [CMP-10 – Assessing competence – evidence, rubrics and moderation](#) — how these claims are tested

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the PML-AI seeded competency set, twenty-four items, reproduced here in the published order.

Competency 16 is stored as Benefits realization and is rendered benefits realisation in this British English text.

- [docs/books/pml-ai/COMPETENCY_MAP.md](#) — the PML-AI competency map, which records that all twenty-four published competencies are covered and that no examination blueprint exists.
- PCI Body of Knowledge domains 8, 9, 10, 11, 12 and 13 ([docs/bok/](#)) — cited by domain, not reproduced.

— Status and version

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CMP-06 · COMPETENCY FRAMEWORKS

Cost and commercial competencies in depth

The chain from estimate to final account, and where competence actually fails along it



VERSION 1.0 · DRAFT

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Cost and commercial competence is not a set of separate skills but one chain — estimate, budget, commitment, accrual, actual, forecast, change, final account — in which each link must reconcile to the last. This document takes the cost and commercial competencies that appear across all three PCI credentials, sets out what distinguishes a practitioner from a professional at each link, and shows with a worked example how a single error at one link is amplified by the time it reaches the forecast.

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Cost and commercial competencies in depth

In one paragraph. Cost and commercial competence is not a set of separate skills but one chain — estimate, budget, commitment, accrual, actual cost, forecast, change, final account — in which each link must reconcile to the last. This document takes the cost and commercial competencies that appear across all three PCI credentials, sets out what distinguishes a practitioner from a professional at each link, and shows with a worked example how a single error at one link is amplified by the time it reaches the forecast. After reading it you should be able to identify which link in your own function is weakest and what evidence would prove it.

Who this is for. Cost engineers, cost managers, quantity surveyors, commercial managers and project controls leads; project finance analysts whose models depend on a construction cost position; and anyone assessing competence claims in this cluster.

— 1. Why the cluster is one chain and not six skills

Cost and commercial work is organised in most businesses as separate roles: estimating, cost control, commercial management, planning, finance. The competencies are assessed separately too. But the numbers they produce are sequential, and each one inherits every error made upstream of it:

estimate → budget → commitment → accrual → actual cost → forecast → change → final account

A person can be flawless at their own link and still be part of a function that reports a wrong number, because the failure is at a handoff. In practice the handoffs are where competence separates, and they are where assessment should look:

Handoff	The question competence answers	The failure when it is absent
Estimate → budget	What was in the estimate that is not in the budget, and who agreed to absorb it?	Budget set below the estimate to fit an approval, with no record of the gap
Budget → commitment	Is the committed value the whole obligation, including options and provisional items?	Commitments understated; the project appears to have contingency it has already spent
Commitment → accrual	What work has been done that has not been invoiced?	Accrual driven by invoices received rather than by progress achieved
Accrual → actual	Does the project cost report reconcile to the ledger, and are the differences explained?	Two cost positions in one organisation, neither owned
Actual → forecast	Which method extrapolates this performance, and what must be true for it to hold?	The tool's default formula, applied to unvalidated inputs
Forecast → change	Is agreed change in the baseline, and unagreed exposure disclosed outside it?	Change buried in the forecast, so the movement cannot be explained
Change → final account	Does the record substantiate the entitlement to someone who was not there?	A defensible entitlement lost for want of a contemporaneous record

The competencies in this cluster, across the three credentials, are: cost management, earned value management, forecasting and EAC, performance measurement and commercial and contract controls (PCL-AI); investment appraisal, financial modelling, capital structure, debt and equity, project contracts and commercial and financial risk (PFL-AI); business cases, procurement, contracts and commercial delivery and cost, schedule and risk integration (PML-AI). They are defined and levelled in CMP-03, CMP-04 and CMP-05. This document goes underneath the definitions.

— 2. Cost management: the accrual is the competency

Most cost control work is mechanical and can be learnt in a quarter. One judgement in it cannot, and it is the accrual: the value of work done that has not yet been invoiced.

A practitioner raises accruals from a defined process — usually a supplier statement, a goods-received record or a progress claim. A professional decides the accrual where those inputs disagree with physical reality, and can defend the number to a finance function that would prefer a smaller one at year end. The specific judgements are:

- **Work done off site.** Fabrication in progress at a supplier's works represents cost incurred and, in most contracts, no payment obligation yet. Whether it is accrued, and on what evidence, is a decision.
- **Disputed valuations.** A contractor claims 400 units of progress; the project assesses 320. The accrual is not the average, and it is not zero because it is disputed. It is the project's honest assessment, with the difference disclosed as exposure.
- **Rework.** Cost incurred on work that will have to be done again is a real cost and a fictional progress. The two entries are separate, and conflating them is how a project reports on budget until the month the rework is discovered.
- **Cut-off.** The same rule, applied on the same date, every period. A month with a late cut-off flatters itself and steals from the next.

The evidence that distinguishes the levels is not the accrual value; it is the substantiation. A signed accrual pack in which each line names its basis — progress record, supplier statement, physical verification, assessed value with the difference disclosed — is professional-level evidence. A spreadsheet of numbers is not, whatever the numbers are.

— 3. Earned value: the competency is validity, not arithmetic

The indices are arithmetic — see BPG-08 — Earned value in practice for the method. The competency question is whether the earned value is true.

Three tests separate the levels, and an assessor can apply all three in ten minutes.

Does the rule of credit reward work or activity? A 0/100 rule on a six-month work package produces five months of nothing and a month of everything, and hides the trend the system exists to reveal. A percentage-complete rule based on the manager's opinion produces steady 8 % per month regardless of what happened. A professional selects the technique per work type — units complete, milestone weighting, level of effort where nothing else is honest — and can say why for each.

Has anyone compared claimed progress to physical progress? The single most valuable half-day in cost control is walking the work with the progress claim in hand. A candidate who has never done this has been operating a system, not validating one.

What happens to level-of-effort scope? Scope measured by time passing always earns exactly as planned and therefore always shows perfect performance. When it is a large share of the baseline, the project's indices are structurally optimistic. A professional knows their level-of-effort share and states it when presenting the indices.

— 4. Forecasting: the method is the competency

An estimate at completion (EAC) is a claim about the future, and every forecasting method encodes an assumption about how the past predicts it:

- **Performance-index extrapolation** ($EAC = BAC \div CPI$, equivalently $BAC \times AC \div EV$) assumes remaining work will be performed at the efficiency achieved so far. Defensible when the remaining work resembles the completed work and the drivers of underperformance persist.
- **Remaining-work re-estimate** ($EAC = AC + \text{a bottom-up estimate of what is left}$) assumes the past performance does not predict the future — typically because the cause was one-off, or because the remaining work is different in kind. Defensible when you can say what changed.
- **Composite approaches** apply different treatments to different scopes: index-based on the like-for-like remainder, re-estimated on the parts that have changed.

Competence is visible in three habits, all of which leave evidence:

1. **The method is chosen and recorded, not inherited.** A dated note naming the method, the alternatives rejected and the reason is professional-level evidence and takes ten minutes to write.
2. **The assumption is stated with the answer.** "EAC 4.4 million, assuming the remaining civils are performed at the current cost performance index of 0.909 and no further scope is instructed" is a forecast. "EAC 4.4 million" is a number.

3. **The forecast is back-tested.** A professional knows whether their own forecasts have historically run low, and by roughly how much. This is the cheapest credibility available to a controls professional and almost nobody does it.

BPG-09 — Estimate at completion owns method selection in full, and TPL-08 provides the scenario comparison sheet.

— 5. Commercial control: the record is the competency

Commercial competence is often assessed by contractual knowledge, which is the wrong test. Contract knowledge is available in the document. What is not available later is the record of what happened.

The professional-level behaviours, stated as artefacts:

- **A notice issued within its period.** Most contracts impose time limits on notification, and an entitlement that was real becomes unrecoverable when a date passes. The competency is a system that triggers the notice, not a memory that recalls the clause.
- **Contemporaneous records that a stranger can use.** Allocation sheets, delay records, photographs with dates and locations, minutes recording instructions. The test: could someone who was not on the project reconstruct entitlement from this file in eighteen months, when the people have left?
- **Exposure kept outside the forecast and visible.** An unagreed variation, a claim in preparation and a potential liability are exposure. Burying them in the forecast destroys the ability to explain a movement; omitting them entirely misleads. The professional reports the forecast and the exposure range separately, and says which is which.
- **A position that concedes what is not defensible.** Competence includes recognising a losing argument early and settling it, rather than carrying a fictional entitlement in the commercial report for a year.

BPG-11 — Change orders and variations and BPG-12 — Claims and extension of time own the mechanics.

— 6. The finance-side members of the cluster

For PFL-AI candidates the same chain runs through different artefacts. The construction cost position feeds the drawdown profile; the contract suite determines whether a cost overrun is borne by the project or by a contractor; and the commercial exposure that a controls function reports outside the forecast is precisely what a lender's technical adviser is looking for. Two boundaries matter for assessment:

- **Financial modelling competence is not cost management competence.** A faultless model implementing an unvalidated cost position produces confident, wrong cover ratios. The competency to interrogate the underlying cost position is a separate one and is the more valuable of the two.
- **Investment appraisal competence is a decision competence.** It is evidenced by the statement of what is being discounted, at what rate and whose return it represents, not by the presence of a net present value in a paper. See CMP-04 §2.4.

— 7. How this goes wrong

The budget is set below the estimate and the gap is not recorded. An approval is granted at a number the estimator did not support, the difference is absorbed silently, and eight months later the overrun is attributed to poor cost control. The competency is the recorded reconciliation between estimate and budget, produced at the time by someone who was willing to have the conversation.

Accruals are driven by invoices. The cost report then measures the supplier's administrative efficiency rather than the project's progress, and it looks best in the months when suppliers are slowest.

Two cost positions coexist. The project cost report and the general ledger differ, both are quoted in different meetings, and nobody owns the reconciliation. The competent response is a monthly reconciliation with each difference explained and aged — not a campaign to make the two systems identical, which will fail.

The forecast method changes without disclosure. A project switches from index-based to a re-estimate in the month the index-based number became unwelcome. The change may be entirely legitimate; not disclosing it is not. State the method every period, and flag the period in which it changed.

Exposure is either buried or omitted. Buried, it destroys the explanation of forecast movement. Omitted, it misleads. Separate reporting of forecast and exposure range is a professional-level habit and the fastest way for a candidate to demonstrate this competency.

Contemporaneous records exist but are unusable. Photographs without dates, allocation sheets without locations, minutes that record attendance rather than instructions. The record is only worth what a stranger can reconstruct from it.

Earned value is validated against the system that produced it. Checking that the report agrees with the tool confirms the export worked. Validation means comparing claimed progress to something outside the system — physical inspection, an independent count, a supplier's own records.

Cost competence is assessed on report quality. The report is the last link and the easiest to make attractive. Assess at the accrual, the reconciliation and the forecast note.

— 8. Worked example: how far one accrual error travels

Illustrative figures. A single control account, currency-neutral cost units (cu), status at month 8 of a 20-month scope. Monetary results rounded to the nearest 1,000 cu; indices shown to three decimal places but computed from unrounded values.

Reported position at month 8

Item	Value (cu)
Budget at completion (BAC)	4,000,000
Planned value to date (PV)	2,400,000
Earned value to date (EV)	2,200,000
Actual cost to date (AC)	2,420,000

Cost performance index: $CPI = EV \div AC = 2,200,000 \div 2,420,000 = 0.909$ Schedule performance index: $SPI = EV \div PV = 2,200,000 \div 2,400,000 = 0.917$ Cost variance: $CV = EV - AC = 2,200,000 - 2,420,000 = -220,000$

Performance-index forecast, computed without rounding by using the equivalent form $EAC = BAC \times AC \div EV$:

$EAC = 4,000,000 \times 2,420,000 \div 2,200,000 = 4,000,000 \times 1.100 = 4,400,000$ $VAC = BAC - EAC = 4,000,000 - 4,400,000 = -400,000$

The commercial layer. Three items sit outside the baseline: variations agreed at 150,000 but not yet incorporated; two instructed but unpriced variations assessed by the project at 90,000, against the contractor's submitted 140,000; and an intimated extension-of-time claim with no cost quantification.

Incorporating only the agreed variations gives a revised BAC of $4,000,000 + 150,000 = 4,150,000$. The forecast now depends on an assumption that must be stated:

- If the variation work is assumed to perform to plan: $EAC = 4,400,000 + 150,000 = 4,550,000$
- If it is assumed to perform at the current efficiency: $EAC = 4,150,000 \times 2,420,000 \div 2,200,000 = 4,150,000 \times 1.100 = 4,565,000$

The difference is $4,565,000 - 4,550,000 = 15,000$, which is simply $150,000 \times 0.100$. Small here, and not small when the variation value is a quarter of the budget. **Reporting either number without naming the assumption is the failure.**

Taking the second treatment as the reported forecast: $VAC = 4,150,000 - 4,565,000 = -415,000$

The unpriced variations are **not** in that number. They are disclosed as exposure of 90,000 to 140,000, unresolved, plus an unquantified time-related cost. Assumptions the answer depends on: no further scope instructed; the current efficiency persists across the remaining work; the disputed valuation is settled at the project's assessment.

Now the error. Suppose the month-8 accrual was understated by 60,000 — one supplier's off-site fabrication omitted because no invoice had arrived. The true actual cost is $2,420,000 + 60,000 = 2,480,000$.

$CPI = 2,200,000 \div 2,480,000 = 0.887$ $EAC = 4,150,000 \times 2,480,000 \div 2,200,000 = 4,150,000 \times 1.127 = 4,678,000$ (to the nearest 1,000 cu)

The forecast moves by $4,678,000 - 4,565,000 = 113,000$ — roughly **1.9 times** the size of the original error. That multiple is not a coincidence: an index-based forecast multiplies an error in actual cost by $\text{revised BAC} \div EV$, here $4,150,000 \div 2,200,000 = 1.886$, and $60,000 \times 1.886 = 113,000$ to the nearest 1,000 cu.

The point for competence assessment: the person who owns the accrual is usually junior, the accrual is usually the least reviewed number in the pack, and the forecast that the board argues about

for an hour inherits its error amplified. This is why cost management competence is assessed at the accrual and not at the report.

— 9. Checklist

Take this into a cost and commercial review, or into an assessment of evidence in this cluster.

- [] Is there a written reconciliation between the estimate and the approved budget, with the gap named?
- [] Are commitments complete, including options, provisional items and unpriced instructions?
- [] Does each accrual line name its basis, and can any line be traced to physical evidence?
- [] Is the cut-off date the same as last period, and is adherence recorded?
- [] Does the project cost report reconcile to the ledger, with differences explained and aged?
- [] What proportion of the baseline is level-of-effort scope, and is that stated when indices are shown?
- [] Has anyone compared claimed progress to physical progress this period?
- [] Does the forecast note state the method, the alternatives rejected and the assumptions?
- [] If the method changed this period, is the change disclosed?
- [] Are agreed changes in the baseline and unagreed exposure reported separately as a range?
- [] Have all notices due under the contract been issued within their periods?
- [] Could a stranger substantiate the current entitlement position from the record file alone?

— Related

- [CMP-03 – PCL-AI: the fourteen competencies](#) — the definitions and levels for the controls-side members of this cluster
- [CMP-04 – PFL-AI: the nineteen competencies](#) — the finance-side members
- [CMP-05 – PML-AI: the twenty-four competencies](#) — procurement, contracts and commercial delivery
- [BPG-09 – Estimate at completion – choosing and defending a method](#) — the forecasting methods in full
- [BPG-11 – Change orders and variations](#) — the change mechanics behind \$5
- [TPL-08 – Estimate at completion scenario comparison](#) — the working sheet for \$4

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the seeded competency sets from which this cluster is drawn.
- PCI Body of Knowledge domains 3, 5, 6, 7 and 11 ([docs/bok/](#)) — budgeting and forecasting, cost management, earned value, contracts and commercial management, and business process cycles. Cited by domain; not reproduced.
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CMP-07 · COMPETENCY FRAMEWORKS

Planning, scheduling and risk competencies in depth

What separates building a schedule from owning one, and maintaining a register from analysing risk

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2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
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— Summary

Planning and risk are the two competencies most often evidenced by artefacts that prove nothing — a large schedule and a tidy register. This document sets out what distinguishes practitioner from professional in each: the four questions a schedule owner must be able to answer, why deterministic dates are systematically optimistic when paths merge, and why an expected-value contingency is exceeded far more often than people expect. Worked figures are included and independently recomputed.

— About this document

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Planning, scheduling and risk competencies in depth

In one paragraph. Planning and risk are the two competencies most often evidenced by artefacts that prove nothing — a large schedule and a tidy register. This document sets out what actually distinguishes a practitioner from a professional in each: the four questions a schedule owner must be able to answer, the structural reason deterministic dates are optimistic when paths merge, and why an expected-value contingency is exceeded far more often than the arithmetic seems to suggest. The figures are illustrative and every one is recomputed from first principles.

Who this is for. Planners, planning managers, risk managers and project controls leads; project managers who commission schedule and risk analysis and have to decide whether to believe it; and assessors reviewing evidence in this cluster.

— 1. What is in the cluster

Across the three PCI credentials this cluster comprises planning and scheduling, project risk and performance measurement (PCL-AI); planning and execution, cost, schedule and risk integration, predictive delivery, agile delivery and hybrid delivery (PML-AI); and commercial and financial risk in its construction-period aspect (PFL-AI). Definitions and level descriptors are in CMP-03, CMP-04 and CMP-05; this document goes underneath them.

Acronyms used here: critical path method (CPM), quantitative schedule risk analysis (QSRA), programme evaluation and review technique (PERT), extension of time (EOT).

— 2. The four questions that separate a schedule owner from a schedule builder

Building a network is a skill that a capable person acquires in months. Owning a schedule is a different competency, and it is testable in a fifteen-minute conversation with four questions.

"What is driving the completion date, and why?" The competent answer names activities, states the float position of the next-nearest paths, and explains what makes this path critical rather than merely long. The incompetent answer names a software view. The follow-up that finds the truth is: what is the second critical path, and how much float separates them? A path with three days of float is not a comfort; it is a second critical path waiting for a delivery to slip.

"Which of these dates are logic-driven and which are constrained?" Every hard constraint in a schedule is a decision that overrides logic, and each one hides float, distorts the critical path, and converts a plan into a target. A professional can list the constraints in their schedule, justify each, and say what the schedule would show without them. Where a constraint is really an instruction — a date the client requires — the professional says so rather than allowing the network to imply it is achievable.

"What is the difference between this plan and the target?" A plan is what will happen if things go as assumed; a target is what someone requires. They are frequently the same document, which is how a project spends a year reporting to a plan nobody believes. The competency is the willingness to maintain and show both, and to state the gap as a number.

"What would change this date?" The answer should be specific and short: a permit, a long-lead delivery, an approval, a labour ramp. If the answer is a general statement about productivity, the schedule is not being used to manage anything.

— 3. Logic quality: the checkable part

Schedule quality has an objective component, and it is the part an assessor can verify without knowing the project. BPG-05 — Schedule quality: a practical review owns the full review; the competency-relevant signals are:

- **Open ends.** Activities without a predecessor or successor float free of the network and cannot drive or be driven. A schedule with many of them is a list, not a model.
- **Excessive lags, especially negative ones.** A lag encodes a duration nobody has named or resourced. Negative lags create logical impossibilities that the software will nonetheless calculate.
- **Constraint density.** Counted, categorised and justified — or unexamined.
- **Negative float.** Its presence means the schedule is asserting the impossible. The competent response is to fix the plan or state the impossibility, not to reduce durations until the number disappears.
- **Duration distribution.** A network in which most activities are exactly 5, 10 or 20 days has been estimated by convention.
- **Resource credibility.** A peak labour requirement that the site cannot physically accommodate makes every date after it fictional.

A candidate whose evidence includes a schedule quality review they performed on someone else's programme, with findings and the argument they had about them, is demonstrating this competency directly.

— 4. Estimating durations: the assumption inside the number

Duration estimates are usually presented as single values, which conceals the assumption that produced them. Where a three-point estimate is used, the arithmetic is trivial and the assumption is not: the formula chosen expresses a belief about the shape of the distribution, and different reasonable formulae give different answers from the same three inputs. §8 shows the arithmetic and the size of the difference.

Two competency markers follow. First, the professional states which formula was used and why, because the choice moves the answer. Second, the professional knows that a three-point estimate on each activity does **not** produce a three-point view of the project, because durations combine — which is the subject of the next section.

— 5. Merge bias: why deterministic schedules are optimistic by construction

When several independent paths feed one milestone, the milestone waits for the slowest of them. Even if every path is equally likely to be early or late, the probability that all of them arrive on time is the product of their individual probabilities, and that product falls fast. §8 works the numbers: three paths, each with a 50 % chance of meeting its date, give the milestone a 12.5 % chance if the paths are independent.

Three consequences a professional acts on:

1. **Deterministic completion dates at merge points are not "the expected date"; they are an optimistic case.** This is arithmetic, not pessimism, and it is why quantitative schedule risk analysis exists.
2. **Correlation matters enormously and is the analyst's hardest judgement.** If the paths share a cause — the same labour pool, the same weather window, the same approving authority — they move together, and the merge penalty shrinks toward nothing. A QSRA run with everything independent overstates the penalty; a QSRA run with correlation assumed away because the software defaults that way is unexamined. Stating the correlation assumption is a hallmark of professional-level competence.
3. **Adding parallel paths into a milestone increases risk even when each path is individually comfortable.** Recovery plans that parallelise work are trading one risk for another, and the trade should be stated.

— 6. Risk competence: register, analysis, ownership, contingency

The register is the artefact and the least informative part. Four things distinguish the levels.

Whether the register contains the risks that matter. The most consequential risks on most projects are the ones with a political cost to writing down: the approval that will not arrive, the client decision that has been outstanding for four months, the design that is not as complete as reported. A register full of weather, labour availability and supply chain generalities has been populated, not compiled.

Whether ownership is real. A risk owned by "the project" is unowned. A risk owned by someone with no authority over its cause is misallocated. The test is whether the owner can name the action they are taking and the date it completes.

Whether the analysis says anything that changes a decision. Probability-times-impact scoring on a five-by-five grid produces a heat map; it does not produce a contingency, and it cannot be aggregated meaningfully because ordinal scores are not numbers. Quantitative analysis, done properly, answers a question a decision-maker has: how much contingency, for what confidence, and against what.

Whether contingency is connected to the analysis and drawn down as risk retires. Contingency set by percentage rule of thumb and never reduced is a budget line, not a risk response. A professional can show the link: this contingency, at this confidence level, against these risks; and as this risk expires, this much is released.

— 7. How this goes wrong

The schedule is maintained but not owned. Updates are applied, the critical path moves, and nobody narrates it. Six months later the completion date has drifted by seven weeks in increments that were never individually significant enough to escalate.

Constraints replace logic. A required date is entered as a hard constraint, the network then reports achievability, and the project reports green until the constrained activity's predecessors physically run out of time. The constraint said what the client wanted; the schedule was never asked what was possible.

Progress updating is confused with re-planning. Applying actual dates and remaining durations updates the record. If the logic that was assumed is no longer how the work will be done, the update produces a precise forecast from an obsolete model.

Duration compression by instruction. Durations are reduced until the completion date fits, without a change to method, resource or scope. Every subsequent forecast inherits the fiction, and the schedule stops being usable as evidence in any later EOT discussion — which is a commercial loss as well as a control one.

The register is audited for completeness rather than content. All fields populated, all dates current, all owners named — and the three risks that will actually decide the outcome are absent because writing them down would have been awkward.

Quantitative analysis is run with default assumptions. Distributions left at the tool's default, correlation left at zero, and the resulting P80 number reported to two decimal places. The output is precise, unexamined and worse than a considered judgement, because it carries unearned authority.

Expected value is treated as a budget. The expected value of a risk set is a long-run average across many repetitions of the same project. There is only one project. §8 shows a case where the expected value is 310,000 cost units, the chance of exceeding it is 37 %, and 80 % confidence needs 550,000.

Contingency is held at the top and never released. Nobody can say what it is for, so nobody can say when it is no longer needed, so it is spent at the end. Tie contingency to named risks at the start and the release becomes a decision rather than an accident.

Agile delivery is assessed with predictive evidence, or the reverse. Empirical throughput data is the evidence for agile competence; baseline integrity and change control are the evidence for predictive competence. Neither substitutes for the other, and CMP-05 §3.6–3.8 sets out the boundary.

— 8. Worked example

Illustrative figures. Three short calculations, each independently recomputed. Durations in working days; costs in currency-neutral cost units (cu). Probabilities are stated as decimals and assumed independent where said; the independence assumption is itself discussed below.

8.1 A three-point duration estimate

An activity is estimated at optimistic 20 days, most likely 30 days, pessimistic 55 days.

Beta (PERT) approximation: $te = (o + 4m + p) \div 6 = (20 + (4 \times 30) + 55) \div 6 = (20 + 120 + 55) \div 6 = 195 \div 6 = 32.5$ days

Triangular mean: $te = (o + m + p) \div 3 = (20 + 30 + 55) \div 3 = 105 \div 3 = 35.0$ days

Standard deviation, beta approximation: $\sigma = (p - o) \div 6 = (55 - 20) \div 6 = 35 \div 6 = 5.83$ days

The two means differ by **2.5 days on a 30-day activity** — 8.3 % of the most likely duration — purely because of the distribution assumed. Neither is more correct in the abstract; the beta form gives the most likely value more weight, which suits work whose upside is limited and whose downside has a long tail. The competency is stating which was used and why, not preferring one.

8.2 Merge bias at a milestone

Three paths feed one milestone. Each path's deterministic date is its median: a 50 % chance of arriving on or before it.

If the paths are independent: $P(\text{all three on time}) = 0.50 \times 0.50 \times 0.50 = 0.125 \rightarrow 12.5 \%$

Two paths: $0.50 \times 0.50 = 0.25 \rightarrow 25 \%$. Four paths: $0.50^4 = 0.0625 \rightarrow 6.25 \%$.

If instead the paths are perfectly correlated — they share a single dominant cause — they all arrive together and the probability returns to 50 %. Real projects lie between these bounds, so the honest statement is: **the milestone's probability is between 12.5 % and 50 %, and where it sits depends on a correlation judgement that must be made explicitly.** Assumptions: each path's distribution is symmetric about its deterministic date, and no path can recover another's delay.

8.3 Expected value is not a contingency

Three cost risks, with single-point impacts, assumed independent:

Risk	Probability	Impact (cu)	Expected value (cu)
A	0.30	400,000	$0.30 \times 400,000 = 120,000$
B	0.60	150,000	$0.60 \times 150,000 = 90,000$
C	0.10	1,000,000	$0.10 \times 1,000,000 = 100,000$
Total			310,000

Total expected value: $120,000 + 90,000 + 100,000 = 310,000$ cu.

But no outcome is 310,000. The eight possible outcomes and their probabilities:

Outcome	Cost (cu)	Probability	Working
None occurs	0	0.252	$0.70 \times 0.40 \times 0.90$
B only	150,000	0.378	$0.70 \times 0.60 \times 0.90$
A only	400,000	0.108	$0.30 \times 0.40 \times 0.90$
A and B	550,000	0.162	$0.30 \times 0.60 \times 0.90$
C only	1,000,000	0.028	$0.70 \times 0.40 \times 0.10$
B and C	1,150,000	0.042	$0.70 \times 0.60 \times 0.10$
A and C	1,400,000	0.012	$0.30 \times 0.40 \times 0.10$
A, B and C	1,550,000	0.018	$0.30 \times 0.60 \times 0.10$

The probabilities sum to 1.000 ($0.252 + 0.378 + 0.108 + 0.162 + 0.028 + 0.042 + 0.012 + 0.018 = 1.000$), which is the check that the enumeration is complete.

Three readings a decision-maker needs:

- **Probability of exceeding the expected value of 310,000:** every outcome above it — $0.108 + 0.162 + 0.028 + 0.042 + 0.012 + 0.018 = 0.370 \rightarrow 37.0\%$. A contingency set at the expected value is exceeded more than one time in three.
- **Probability of spending nothing:** 25.2 %. The single most likely outcome is B only at 150,000 (37.8 %) — neither of which is anywhere near the average.
- **Contingency for 80 % confidence:** accumulating the sorted outcomes gives 0.252 at 0 cu, 0.630 at 150,000, 0.738 at 400,000 and **0.900 at 550,000**. The first level reaching at least 80 % confidence is therefore **550,000 cu** — $550,000 \div 310,000 = 1.77$ times the expected value.

Assumptions this depends on: three risks only; independence; single-point impacts rather than ranges; no correlation between A, B and C; and no risk not on the register. Every one of those assumptions is generous, which is why real quantitative analysis uses simulation over impact ranges with stated correlation — and why the number it produces is a decision input, not an answer.

9. Checklist

For a schedule and risk review, or for assessing evidence in this cluster.

Schedule:

- [] Can the owner name the current critical path, the second path, and the float between them?
- [] Are all hard constraints listed, justified, and their effect on float understood?
- [] Is there a stated difference between the plan and the required target date?
- [] Is there a basis-of-schedule document naming the assumptions, calendars and productivity basis?
- [] Open ends, negative lags, negative float and constraint counts — measured this period?
- [] Do peak resource requirements fit the physical and contractual reality of the site?
- [] Was the last update a progress update, a re-plan, or a re-plan presented as an update?

Risk:

- [] Which risks are missing because writing them down would be awkward?
- [] Does every risk owner have authority over the cause, and a dated action?
- [] Is the analysis quantitative where the decision needs a number, and qualitative only where it does not?
- [] Are correlation assumptions stated explicitly rather than left at the tool's default?
- [] Is contingency tied to named risks, at a stated confidence level?
- [] Is contingency released as risks expire, with a record of each release?
- [] Does anyone present expected value as though it were a budget?

— Related

- [CMP-03 – PCL-AI: the fourteen competencies](#) — planning and scheduling, project risk and performance measurement, defined and levelled
- [CMP-05 – PML-AI: the twenty-four competencies](#) — planning and execution, integration, and the three delivery approaches
- [BPG-05 – Schedule quality – a practical review](#) — the full review method behind §3
- [BPG-10 – Contingency and management reserve](#) — how contingency is set, held and released
- [BPG-17 – Quantitative schedule risk analysis](#) — the analysis method behind §5 and §8.2
- [TPL-11 – Quantitative schedule risk analysis input sheet](#) — the input instrument

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the seeded competency sets from which this cluster is drawn.
- PCI Body of Knowledge domains 6, 8, 9, 10 and 12 ([docs/bok/](#)) — earned value and forecasting, project lifecycle, adaptive delivery, scheduling and risk management. Cited by domain; not reproduced.
- The three-point estimating approximations in §8.1 are long-standing published techniques described in our own words; no standard's text or tables are reproduced. All figures in §8 are illustrative and recomputed from first principles.

— Status and version

Founding-stage document · Version 1.0 — effective date to be confirmed · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

CMP-08 · COMPETENCY FRAMEWORKS

Data, digital and AI competencies in depth

The chain from data readiness to human accountability, and the evidence that proves each link

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

All three PCI credentials carry a group of data, digital and AI competencies, and in every case they form one chain: data readiness, analysis, workflow design, reporting and automation, governance, and the human validation or accountability that closes it. This document sets out what each link requires, what distinguishes practitioner from professional, and how each is evidenced by a decision rather than by a tool. It includes a worked back-test showing how to state a model's error and its bias.

— About this document

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Data, digital and AI competencies in depth

In one paragraph. All three PCI credentials carry a group of data, digital and AI competencies, and in every case they form one chain: data readiness, analysis, workflow design, reporting and automation, governance, and the human validation or accountability that closes it. This document sets out what each link requires, what distinguishes a practitioner from a professional, and how each link is evidenced by a decision rather than by a tool. It includes a worked back-test showing how to state a model's typical error and its bias, and what to do with each.

Who this is for. Controls, finance and delivery professionals who use AI-assisted methods in work that someone relies on; heads of function deciding what to admit into their process; and assessors reviewing evidence in this cluster.

— 1. The chain, and where each credential's competencies sit on it

The Institute's position is a single sentence, and everything below is its operating detail: **AI proposes; the professional disposes.** An output is fit to be relied upon when it is explainable, validated and owned by a competent human.

The competencies that carry that position are distributed across the three credentials, but they occupy the same six positions:

Link	PCL-AI	PFL-AI	PML-AI
Data foundation	implicit in predictive analytics	implicit in predictive cash-flow analysis	implicit in digital delivery
Analysis	Predictive analytics	Predictive cash-flow analysis; digital due diligence	Decision intelligence
Workflow design	AI-enabled project controls	AI-enabled financial modelling	AI-enabled project management
Carriage	Digital reporting; automation	within digital due diligence	Digital delivery; automation
Governance	Responsible AI	Responsible AI	Responsible AI
Closure	Human validation	Human validation	Human accountability

Two observations that matter for assessment. First, the data foundation is not itself a named competency in any of the three sets — it sits inside the analysis competencies, which means an assessor has to probe for it deliberately or it goes unexamined. Second, the closing link differs: PCL-AI and PFL-AI name human validation (the act of checking), while PML-AI names human accountability (answerability for the outcome, including for checks performed by others). The distinction is real and is drawn in §7.

AIG-01 to AIG-12 own the Institute's teaching on AI in a controls function. This document is about what competence in it looks like and how it is proved.

— 2. Data readiness, and the question that exposes it

Every analytical or AI-assisted method inherits the quality of what it was given. The professional-level marker is the ability to state, before running anything, what the available data can and cannot support.

The four properties that decide it:

- **Consistency of definition over time.** A cost series in which the coding structure changed eighteen months ago is two series wearing one name. Any trend across the join is an artefact.
- **Completeness of the periods that matter.** Missing data is rarely random. The periods most often missing are the disrupted ones, which are exactly the periods a risk or forecast model most needs.
- **Granularity relative to the question.** Monthly totals cannot answer a question about weekly productivity, however sophisticated the method applied.
- **Independence from the answer.** Data that was itself produced by a process the answer will judge — progress claimed by the party being assessed — carries the incentive with it.

The evidence is a document in which someone declined or reshaped a request: "the question as asked cannot be answered from this data; here is what it can support, and here is what would have to be collected to answer the original question." That artefact is rare and it is nearly conclusive of professional-level competence. AIG-03 — Data readiness covers the underlying assessment.

— 3. Analysis: the competency is the error statement, not the number

Anyone can produce a forecast. The competency is producing one accompanied by an honest account of how wrong it usually is.

Three habits distinguish the levels:

Hold-out testing. Fit on part of the history, test on the part withheld, report the result. A model evaluated only on the data it was fitted to has been described, not tested.

Separating noise from bias. These are different problems with different remedies. Noise — a model that is wrong by a similar amount in both directions — is reduced by better data or a better method. Bias — a model that is consistently wrong in one direction — can be corrected arithmetically once measured, and usually points to something structural. §9 works both numbers from the same four observations.

Stating the range and its basis. A single-point analytical output invites the reader to treat it as precise. A range with a stated basis invites them to treat it as a forecast, which it is.

The recurring false positive is high explanatory power over history. A model that fits the past closely is easy to build and easy to sell; whether it predicts anything is a separate question, answered only out of sample. A candidate who has reported that their own model was wrong, and by how much, is demonstrating something a candidate who reports only successes cannot.

— 4. Workflow design: where the output enters, and what stands between it and reliance

This is the operating-model competency — AI-enabled project controls, AI-enabled financial modelling, AI-enabled project management — and it is the one most often evidenced with a list of tools, which answers a question nobody asked.

Designing an AI-assisted step means specifying six things, and the specification is the evidence:

1. **The task**, narrowly. "Summarise the 40 supplier reports into an exceptions list against these five criteria" is a task. "Help with reporting" is not.
2. **The input contract** — what the model receives, in what form, and what it must never receive (personal data, priced commercial positions, counterparty confidential material).
3. **The output contract** — the form the output must take, so that it can be checked mechanically where possible. Output that must be read to be verified costs more to check than to produce.
4. **The entry point** — precisely where in the process the output is used, and by whom.
5. **The check** — what stands between the output and reliance, described in §7.
6. **The failure mode** — what a wrong output looks like here, and what it would cause. Some tasks fail loudly and cheaply; others fail silently into a board paper.

The decisive professional-level marker is an **exclusion**: a task deliberately not delegated, with the reason recorded. Every competent design has some. A workflow with no exclusions has not been designed; it has been adopted.

— 5. Carriage: reporting and automation, and the problem of silent failure

Digital reporting and automation carry the output to the point of use, and they share one failure mode that separates the levels: **failing quietly while continuing to produce plausible numbers**.

The competency is expressed in build decisions:

- **Lineage**. Every figure in a report traces to a source system, a field and an extraction time. Where it cannot, that figure is manual and should be labelled as such.
- **Reconciliation as a control, not an activity**. An automated pipeline that reconciles its own totals to the source and stops on a difference is a control. One that reconciles when someone remembers is not.
- **Break-detection sized to the error that matters**. A row-count check catches a truncated extract; it does not catch a mapping error that silently reassigns cost to the wrong account. Design the check for the failure that would be expensive, not the failure that is easy to detect.
- **Survivability**. Documented, owned, version-controlled, and demonstrably operable by someone who did not build it. An automation that only its author can maintain has converted a manual process into a single-person dependency and called it efficiency.

The assessment question is one sentence: when this breaks, how will anyone know? If the honest answer is "when someone notices the number looks odd", the automation is not at professional level however elegant its construction.

— 6. Governance: responsible AI as a decision, not a document

Responsible AI appears in all three sets. Its content is well covered in AIG-08 — Governing AI on a project and AIG-09 — Bias, explainability and auditability; what matters here is what evidences it.

Four decisions constitute the competency:

- **Data boundaries.** Which categories of information may be entered into which class of tool. The boundary is only real if someone has been told no.
- **Disclosure.** Where AI assistance is disclosed to a reader whose reliance would change if they knew. A convention applied consistently — including to a board paper, and including when it invites an awkward question — is the evidence.
- **Explainability as a precondition.** For some decisions, an unexplainable output is inadmissible regardless of accuracy, because the decision has to be defended to someone entitled to an explanation. Knowing which of your decisions those are is a professional-level judgement.
- **Prohibition.** At least one use case declined. A framework that has never prevented anything is a document.

An open question for this framework, recorded rather than assumed: [CONFIRM: whether AI assistance must be declared in candidate-submitted competence evidence, and in what form].

— 7. Closure: validation, and the difference between validating and being accountable

Human validation (PCL-AI, PFL-AI) is the act: checking an AI-derived or automated output before reliance, and owning it as your own work. A check is capable when four things are true:

1. **It is independent of the thing being checked.** Re-running the same model, or asking the same model whether it is right, is not a check. The independent source may be a different system, a physical observation, a manual recomputation of a sample, or a person who did not see the first answer.
2. **It could actually fail.** If no realistic output would be rejected, the check is a formality. A useful discipline is to specify the tolerance before seeing the output.
3. **It is sized to the risk.** Full recomputation for a board forecast; a stated sample for a routine extraction — with the sample size and selection method recorded, because an unstated sample is an anecdote.
4. **It leaves a record.** What was checked, against what, by whom, on what date, and what was found — including the times nothing was found, because a record showing only clean checks is a record of a process nobody ran.

Human accountability (PML-AI) is broader: being answerable for an outcome, including for work checked by others and produced by systems. It has three components — a named owner for every output that reaches a decision, the refusal to release what you cannot stand behind, and the absence of deflection to "the model said" or "the system produced it". The organisational version is the design obligation: no decision may be taken by a process with no owner. A function where every individual validated correctly and nobody was accountable for the aggregate has an accountability gap by design, and it will be discovered by an incident rather than by an audit.

— 8. How this goes wrong

Tools are offered as evidence of competence. A list of platforms describes a procurement history. Every competency in this cluster is evidenced by a decision: what enters where, what check stands between the output and reliance, what was excluded, what was prohibited.

The check is performed by the same system that produced the output. Asking a model to review its own work produces agreement, which is not verification. The independence requirement in §7 is the whole point.

Validation is a signature. "Reviewed by" with no statement of what was checked against what records that a signature occurred. A single line — "totals reconciled to the ledger extract of 30 June; 10 of 240 line codings sampled, one error found and corrected" — turns it into evidence.

A model's fit is reported as its accuracy. Explanatory power over history is a statement about the past. Out-of-sample error is a statement about the model. Only the second belongs next to a forecast.

Bias is treated as noise. A model that has run 3 % low for two years does not need more data; it needs a correction and an investigation into the structural cause. Reporting only absolute error hides this entirely, which is why §9 computes both.

Automation is measured by hours saved. Hours saved is a benefit claim. The competency question is what happens when it fails, and a silent failure in an automated pipeline can cost more than the automation ever saved.

Disclosure is applied selectively. AI assistance is disclosed in routine documents and omitted from the paper that matters, on the grounds that it would raise questions. That is precisely the document where the reader's reliance would change, and therefore the one where disclosure is owed.

Responsible AI is evidenced by a policy. A policy is a document until it prevents something. Name the prohibited use case and the reasoning.

The data foundation is never examined. Because no credential names data readiness as a standalone competency, it is the easiest link to skip — and it is the link on which every other link in the chain depends. Ask what the data can support before asking what the model concluded.

— 9. Worked example: stating a model's error and its bias

Illustrative figures. A cost forecasting model has produced a monthly forecast for four completed periods. Currency-neutral cost units (cu). Percentage errors are expressed **as a percentage of actual** and shown to two decimal places; money rounded to the nearest 1,000 cu.

Period	Forecast (cu)	Actual (cu)	Error = actual – forecast (cu)	Error as % of actual
1	1,200,000	1,320,000	+120,000	+9.09 %
2	900,000	855,000	–45,000	–5.26 %
3	1,500,000	1,725,000	+225,000	+13.04 %
4	1,100,000	1,067,000	–33,000	–3.09 %

Substitutions for the percentage column: $120,000 \div 1,320,000 = 0.0909 = +9.09\%$ · $-45,000 \div 855,000 = -0.0526 = -5.26\%$ · $225,000 \div 1,725,000 = 0.1304 = +13.04\%$ · $-33,000 \div 1,067,000 = -0.0309 = -3.09\%$

Typical size of error — mean absolute percentage error (MAPE), the average of the errors ignoring sign:

$$\text{MAPE} = (9.09 + 5.26 + 13.04 + 3.09) \div 4 = 30.48 \div 4 = 7.62\%$$

Direction of error — mean percentage error, the average keeping the sign:

$$\text{MPE} = (+9.09 - 5.26 + 13.04 - 3.09) \div 4 = 13.78 \div 4 = +3.45\%$$

The two numbers say different things. The model is typically wrong by **7.62 % of actual**, and it is wrong **low** by **3.45 %** on average — so roughly $3.45 \div 7.62 = 45\%$ of its typical error is systematic rather than random. Noise needs a better method or better data; bias needs a correction and an explanation.

Applying it to a live forecast. The model now outputs 2,400,000 cu for the current period. Because the error is measured as a proportion of actual, the bias-corrected central estimate is:

$$\text{corrected} = 2,400,000 \div (1 - 0.0345) = 2,400,000 \div 0.9655 = 2,486,000 \text{ cu (to the nearest 1,000)}$$

— an uplift of $2,486,000 - 2,400,000 = 86,000$ cu. A working band of plus or minus the typical error around that central estimate gives:

$$2,486,000 \times 0.0762 = 189,000 \rightarrow \text{range } 2,297,000 \text{ to } 2,675,000 \text{ cu}$$

What this example does not establish. Four observations cannot demonstrate that a bias exists; the apparent 3.45 % could easily be four periods of ordinary variation. The MAPE band is a working range, not a statistical confidence interval, and it assumes future errors resemble past ones — which fails precisely when a project changes character. The teaching point is the discipline, not the figures: **a forecast presented without its error history is an opinion with decimal places**, and the professional-level behaviour is to keep the back-test, report both numbers, and say how many observations sit behind them.

— 10. Checklist

Before relying on an analytical or AI-assisted output, or when assessing evidence in this cluster:

- [] What question is this answering, and can the underlying data support that question?
- [] Has any definition in the data series changed during the period being analysed?
- [] Was the model tested on data it had not seen?
- [] Are both the typical error and the directional bias reported, and over how many observations?
- [] For each AI-assisted step: task, input contract, output contract, entry point, check, failure mode — all specified?
- [] Which tasks were deliberately **not** delegated, and is the reason recorded?
- [] Is the check independent of the thing being checked, and could it realistically fail?
- [] Is the check sized to the risk, with sample size and selection recorded?
- [] When the automation breaks, how will anyone know — and has that detection ever fired?
- [] Does every figure in the report trace to a source, field and extraction time?

- [] Is AI assistance disclosed to readers whose reliance would change if they knew — including in the documents where the question would be awkward?
- [] Is there a named person accountable for every output that reaches a decision?

— Related

- [CMP-03 – PCL-AI: the fourteen competencies](#) — predictive analytics, AI-enabled project controls, digital reporting, automation, responsible AI, human validation
- [CMP-04 – PFL-AI: the nineteen competencies](#) — AI-enabled financial modelling, predictive cash-flow analysis, digital due diligence
- [CMP-05 – PML-AI: the twenty-four competencies](#) — AI-enabled project management, decision intelligence, digital delivery, human accountability
- [AIG-03 – Data readiness: what AI needs before it is any use](#) — the assessment behind §2
- [AIG-09 – Bias, explainability and auditability](#) — the governance content behind §6
- [AIG-10 – Human in the loop: what AI may and may not decide](#) — the decision boundary behind §7

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the seeded competency sets from which this cluster is drawn.
- PCI Body of Knowledge domain 13, AI for Project Controls and PM: concepts, tools and practice ([docs/bok/](#)) — cited by domain; not reproduced.
- The error measures used in §9 (mean absolute percentage error and mean percentage error) are standard published definitions described in our own words. All figures are illustrative and recomputed from first principles.

— Status and version

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CMP-09 · COMPETENCY FRAMEWORKS

Behavioural and leadership competencies

The behaviours that decide whether a correct number changes anything

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

A correct forecast that nobody acts on has changed nothing. This document sets out the five behaviours that determine whether a controls, finance or delivery professional is listened to — raising a number that is unwelcome, holding a position under pressure and releasing it on evidence, stating what is not known, making a trade-off legible rather than winning an argument, and owning delegated and AI-derived output — with the observable indicators for each, the escalation ladder that supports them, and the organisational conditions without which none of them survives.

— About this document

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Behavioural and leadership competencies

In one paragraph. A correct forecast that nobody acts on has changed nothing. This document sets out the five behaviours that determine whether a controls, finance or delivery professional is listened to — raising a number that is unwelcome, holding a position under pressure and releasing it on evidence, stating what is not known, making a trade-off legible rather than winning an argument, and owning delegated and AI-derived output — with the observable indicators and counter-indicators for each, the escalation ladder that supports them, and the organisational conditions without which none of them survives.

Who this is for. Project controls managers, cost and planning leads, project and programme managers, and the executives who decide whether these behaviours are rewarded or quietly penalised in their organisations.

— 1. Behaviour is not a separate framework

The Institute does not publish a behavioural competency framework alongside the technical ones, and the omission is deliberate. Behaviour is inside the published sets, because it is inseparable from the technical work.

It appears explicitly. PML-AI names project leadership, stakeholder leadership, governance and human accountability. PCL-AI names project controls governance and human validation; PFL-AI names human validation. These are behavioural competencies with technical content, not soft additions.

It also appears implicitly, in every level descriptor at professional and above. CMP-02 §2.3 defines professional level as deciding the method where procedure is silent **and defending the decision to someone who wanted a different answer**. That second clause is a behaviour. The scale cannot be climbed by technical capability alone, because the level is defined partly by conduct under pressure.

There is a structural reason this matters more in project controls, project finance and delivery management than in many disciplines. The output of the work is a number that constrains other people's plans, budgets, bonuses and reputations. The number is produced by someone who usually does not hold authority over the project, is often junior to the people it constrains, and frequently has to deliver it to the person whose performance it describes. That configuration puts consistent pressure in one direction: the forecast is asked to be lower, the risk to be smaller, the date to be earlier. Nobody has ever been asked to make a forecast worse.

— 2. The five behaviours

Each is stated with observable indicators, counter-indicators, and the evidence that would demonstrate it. Indicators are observable actions, not attitudes; nobody can assess "integrity", and everybody can assess whether a forecast moved before or after the client found out.

2.1 Raising a number that will be unwelcome

What it is. Reporting a deterioration at the point you are confident of it, rather than at the point it becomes undeniable — while there is still time for the information to be useful.

Observable indicators. The forecast moved in a period when no external event forced it. The escalation is dated earlier than the event that confirmed it. The person raised the issue in a written record, not only in a corridor. They brought the size of the problem and its basis, not a warning that things "may be tight". They said what they wanted done about it.

Counter-indicators. Forecasts that move only in the month a client audit is announced. Deterioration disclosed in a footnote or a verbal aside. A record showing the person "flagged concerns" without a number or a date. A pattern of large single-period corrections, which is the signature of accumulated withholding.

Evidence. A dated forecast note showing an upward revision and its basis, together with the outcome. A report qualification in writing. An escalation email with its date, and the subsequent record of what happened.

2.2 Holding a position under pressure — and releasing it on evidence

What it is. Two behaviours that only count together. Maintaining a position when it is challenged by someone with more authority and no new evidence; and changing it promptly and publicly when new evidence arrives.

Observable indicators. The person can state, in advance, what would change their mind — a specific piece of evidence, not a threshold of pressure. When challenged, they ask what is being asserted rather than restating their number louder. When they concede, they say what changed and update the record. When they do not concede, they offer a route to resolve it: an independent check, a sample verification, a joint walk of the work.

Counter-indicators. Positions that change after a meeting with a senior person and no new information, with no record of why. Positions that never change, which is not integrity but a different failure — the person has stopped processing evidence. Conceding privately while maintaining the original position in the record, or the reverse.

Evidence. A challenge and its resolution, documented on both sides. A position reversed with the reason recorded contemporaneously. A dispute escalated to an independent check, and the outcome.

2.3 Saying what you do not know

What it is. Distinguishing, out loud and in the record, between what has been verified, what has been assumed, and what is unknown — and saying what it would take to know it.

Observable indicators. Reports carry an explicit assumptions section that changes between periods rather than being copied forward. The person answers a question with "I don't know; I can find out by Thursday" and then does. Ranges are used where the situation warrants a range, with the drivers of the spread named. Unverified inputs are labelled as unverified in the document, not just in the author's mind.

Counter-indicators. Confident single-point answers to questions the data cannot support. An assumptions section identical to last quarter's. Precision used as a proxy for confidence — a forecast quoted to the nearest unit when the underlying accrual is an estimate.

Evidence. Reports with live assumption registers. A documented refusal to provide a number on inadequate data, with the alternative offered and the date it was delivered. A stated range with its basis, later compared to outturn.

2.4 Making the trade-off legible rather than winning the argument

What it is. Presenting a decision-maker with the actual choice and its consequences, in a form they can decide from, rather than advocating for the answer you prefer and treating disagreement as a failure of theirs.

Observable indicators. Options are presented with consequences on the dimensions the decision-maker cares about — cost, date, scope, risk exposure — not only the one the analyst cares about. The recommendation is stated and separated from the analysis, so the decision-maker can accept the analysis and reject the recommendation. The person can articulate the strongest version of the option they are advising against. After the decision, they implement it and record any residual concern once, rather than relitigating it monthly.

Counter-indicators. One real option and two straw men. Analysis that stops precisely where it would have become inconvenient. Escalation of a decision that has already been properly made, in the hope of a different answer. A residual concern raised for the first time after the outcome is known.

Evidence. An options paper with genuine alternatives and comparable consequence statements. A decision record showing the recommendation, the decision taken and any recorded dissent. A decision you disagreed with, implemented, and whose consequences you tracked honestly.

2.5 Owning the output, including what a system produced

What it is. Being the person answerable for a number, including when the analysis came from a model, a tool or a team — and declining to release what you cannot stand behind.

Observable indicators. The person can explain a figure produced by an AI-assisted or automated process in their own words, including its limitations. They can name what they checked and against what. They have declined to issue something within the reporting deadline because it could not be verified, and said so rather than issuing it silently. They do not attribute an error to a tool or to a junior colleague in a forum where neither is present.

Counter-indicators. "The model produced it." "That comes from the system." A signature on a document whose method its signatory cannot describe. Errors traced downward in public and credit traced upward in private.

Evidence. A validation record naming what was checked and found. A withheld or qualified deliverable with the reason recorded. A post-incident review in which the person owned a failure in a process they designed. See CMP-08 §7 for the validation mechanics and CMP-05 §4.9 for accountability at leader level.

— 3. The escalation ladder

Behaviour needs a route, or it becomes a choice between silence and resignation. The ladder below moves only when the previous rung has been tried and recorded, and each rung leaves evidence.

1. **Fix it in the record.** Correct the number, note the correction and its basis. Most issues stop here.
2. **Raise it with the person who owns the number.** Directly, with the basis, and with what you want changed. Confirm the conversation in writing the same day — not as a threat, but because a contemporaneous note is the only version that survives.
3. **Qualify the report in writing.** If a report must be issued and you cannot sign it cleanly, say what you could not verify and what the number excludes. §7 gives the wording. This is the most under-used rung on the ladder and the most powerful, because it is factual, proportionate and permanent.
4. **Escalate to the next governance tier.** The project board, the steering committee, or the function's head — whichever the governance framework names. Bring the number, the basis, the attempts already made, and the specific decision you are asking for.
5. **Escalate to the accountable officer, formally.** In writing, with the record. At this rung the matter has stopped being a disagreement about a number and has become a governance failure.
6. **Treat it as a conduct matter.** Where a number is being misstated knowingly, or where you are being instructed to misstate one, this is no longer a controls question. ETH-01 — The PCI code of ethics and professional conduct and ETH-03 — Reporting numbers under pressure own this ground, including the protections and the obligations.

Two rules govern the ladder. **Proportionality:** starting at rung 4 for a rung-2 problem destroys the credibility you will need later. **Documentation at every rung:** the record is not for a tribunal, it is for the ordinary case where three months later everyone remembers the conversation differently.

— 4. The conditions that make these behaviours possible

An organisation cannot certify a behaviour it punishes, and most organisations punish these behaviours without deciding to. Four conditions determine whether the behaviours in §2 are available:

- **A forecast can move without someone being blamed for the movement.** Where a deteriorating forecast is treated as the forecaster's failure, forecasts stop deteriorating. The observable test: when a forecast worsened last quarter, was the first question what changed? or who allowed this?
- **The controls function reports somewhere other than solely to the person whose performance it measures.** Complete dependence on the project manager for appraisal, allocation and progression makes independent reporting a personal-cost decision every month. A dotted line to a functional head is a structural control, not an organisational preference.
- **There is time to verify.** A reporting cycle that leaves two days between data cut-off and board pack makes physical verification impossible, and then treats its absence as an individual failing.
- **Bad news is received, not punished, in the forum where it is delivered.** People calibrate to what happens to the last person who raised something. Everyone in the room learns from that, and they learn it in one meeting.

For an employer using this framework, the practical consequence is uncomfortable: if you are assessing staff against §2 while running an organisation that fails §4, you are assessing their appetite for personal cost rather than their competence.

— 5. Evidencing behaviour

Behaviour is evidenced badly by testimonial and well by contemporaneous record. Four artefact types carry almost all of it:

- **The dated note.** A forecast note, an escalation email, a file note of a conversation, written the same day. Its value is entirely in its date.
- **The qualification.** A written statement in a deliverable of what could not be verified or what is excluded. It is public, permanent and hard to reinterpret later.
- **The reversal.** A position changed with the reason recorded, which evidences §2.2 more strongly than any number of held positions.
- **The reconciliation over time.** What you said in month 6 set against what happened by month 12. Very few candidates offer this, and it is the single most persuasive behavioural evidence available, in both directions: a professional who was wrong and says so has demonstrated more than one who was right.

A behavioural claim with no contemporaneous artefact is a story. Not necessarily untrue — but not assessable, and CMP-10 §2 explains why an assessor cannot give it weight.

— 6. How this goes wrong

Behaviour is assessed as personality. "Confident communicator", "team player", "strong presence" — none of which is observable in the sense required, and all of which correlate with things a competency framework must not select for. Assess actions with dates.

Escalation is confused with courage. Escalating everything is not a behavioural strength; it is a failure of the judgement in §3, and it exhausts the credibility needed for the escalation that matters.

Holding a position is confused with being right. A person who never changes their view is not reliable; they have stopped updating. The competency is the pair of behaviours in §2.2, and evidence for only one half of it is evidence of the wrong thing.

The qualification is used as a weapon. A report qualified in aggressive or accusatory terms invites a fight about tone and buries the substance. The wording in §7 is deliberately flat: it states what was and was not verified, and nothing about anyone's motives.

Bad news is delivered at the wrong size. A three-week slip presented with the gravity of a project failure trains the audience to discount you; a serious problem mentioned in passing under "any other business" is a disclosure that will not protect anyone, including its author.

The organisation certifies behaviour it structurally prevents. See §4. A behavioural rating scale deployed into an environment that punishes the behaviour produces uniformly high ratings and no information.

Nobody records anything until it is a dispute. By then the record is written by people who know how it turned out, and the version most favourable to its author is the one that gets written. The whole behavioural evidence base depends on notes made before anyone knew the ending.

— 7. Worked example: the wording of a qualification

Illustrative wording; the figures within it are illustrative. A cost report must be issued to a project board on the due date. Two problems remain unresolved: the December progress claim has not been verified against physical progress, and two instructed variations are unagreed. The professional cannot delay the report and cannot sign it cleanly.

The qualification, placed in the report's summary rather than in an appendix:

Basis and limitations of this report. The forecast at completion of 4,565,000 cost units assumes the remaining work is performed at the cost performance index achieved to date (0.909) and that no further scope is instructed. It includes agreed variations of 150,000 cost units. It **excludes** two instructed but unpriced variations, assessed by the project at 90,000 cost units against the contractor's submitted 140,000 cost units, and any cost consequence of the extension-of-time claim intimated on 12 November, which is not yet quantified. The December progress claim has not been verified against physical progress; verification is scheduled for 14 January. If the claim proves overstated, the earned value and therefore the forecast will move adversely.

Four properties make this work, and they are worth copying:

1. **It states the number and its assumptions in the same breath**, so nobody can quote the forecast without the condition attached.
2. **It separates what is included from what is excluded**, and gives the excluded items a range rather than a silence.
3. **It names a date for resolution.** A qualification with no closing action becomes permanent furniture and stops being read.
4. **It attributes nothing to anyone.** No motive, no blame, no adjectives. It is a statement about the report, which makes it very difficult to argue with and very easy to act on.

The consequence of writing it is that the board can no longer treat 4,565,000 as a settled figure — which is the entire point, and the reason the paragraph is harder to write than the model that produced the number.

— 8. Checklist

For a professional deciding how to handle a number nobody wants:

- [] Am I confident of this, or confident that it is likely? Which am I about to say?
- [] Is it early enough for the information to be useful, and if not, why did it take this long?
- [] Do I have the number, the basis and the specific decision I am asking for?
- [] Have I written it down today, with a date?
- [] What evidence would change my mind, and have I said so out loud?
- [] If I must issue this, what exactly can I not verify, and does the report say so in its summary?
- [] Which rung of the escalation ladder is proportionate — and have I tried the one below it?

- [] Am I presenting a decision or advocating a preference? Have I stated the strongest case against my own recommendation?
- [] For any AI-assisted or delegated figure: can I explain it in my own words, and did I check it?
- [] After the decision, will I implement it properly, or relitigate it monthly?

For an employer using this document to assess others: answer §4 first.

— Related

- [CMP-02 – Competency levels and how they are evidenced](#) — why professional level is defined partly by conduct under challenge
- [CMP-05 – PML-AI: the twenty-four competencies](#) — project leadership, stakeholder leadership and human accountability, defined and levelled
- [CMP-10 – Assessing competence – evidence, rubrics and moderation](#) — how behavioural claims are tested without assessing personality
- [ETH-01 – The PCI code of ethics and professional conduct](#) — the obligations behind rung 6 of the ladder
- [ETH-03 – Reporting numbers under pressure – a casebook](#) — the cases this document's behaviours are drawn from
- [CAR-05 – Into project controls leadership](#) — the career transition in which these behaviours become the job

— Sources and standards

- PCI platform certification catalogue ([backend/Data/MultiCert.cs](#)), verified August 2026 — the seeded competency sets, in which the behavioural competencies named in §1 appear.
- PCI Body of Knowledge domains 4 and 13 ([docs/bok/](#)) — performance management and management reporting, and AI for project controls. Cited by domain; not reproduced.
- No case studies, quotations, organisations or individuals are cited in this document. The wording in §7 is an illustrative construction, not a reproduction of any real report.

— Status and version

Founding-stage document · Version 1.0 — effective date to be confirmed · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

CMP-10 · COMPETENCY FRAMEWORKS

Assessing competence — evidence, rubrics and moderation

How a competency claim is tested, why rubrics fail, and what
moderation is actually diagnosing

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

An assessment is an inference from a sample of evidence to a claim about capability, and it can fail in two directions with very different costs. This document sets out the evidence types and what each can and cannot show, how to write a rubric descriptor that two assessors read the same way, the difference between compensatory and conjunctive scoring and when each is legitimate, how moderation works and what its disagreements actually diagnose, and why self-assessment on its own carries no assessment weight.

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Assessing competence — evidence, rubrics and moderation

In one paragraph. An assessment is an inference: from a sample of evidence, to a claim about what a person can do in situations nobody has observed. It can fail in two directions with very different costs. This document sets out the evidence types and what each can and cannot show, how to write a rubric descriptor that two assessors read the same way, the difference between compensatory and conjunctive scoring and when each is legitimate, how moderation works and what its disagreements actually diagnose, and why self-assessment on its own carries no assessment weight — together with the three things it is genuinely good for.

Who this is for. Anyone who assesses competence and has to defend the decision: certification assessors, heads of function running internal competency schemes, and employers whose promotion and deployment decisions rest on a rating someone produced.

— 1. What an assessment is claiming, and how it fails

Every competence decision has the same shape. A person's capability across an entire working life is unobservable. A sample of it is observed — some artefacts, an examination, a conversation — and from that sample an inference is drawn about behaviour in situations that were not sampled.

Two properties determine whether the inference holds.

Validity — is the evidence about the thing being claimed? A written examination on forecasting methods is valid evidence of knowledge of forecasting methods. It is weak evidence of whether the candidate will hold a forecast under commercial pressure, because that situation is not in the sample. Most assessment failures are validity failures wearing the clothes of rigour: the instrument is precise, consistent and measuring something adjacent to the claim.

Reliability — would the same evidence produce the same decision with a different assessor, on a different day? An unreliable process is unfair in a way that is invisible to everyone inside it, because each assessor believes their own judgement to be the standard.

The two failure directions have asymmetric costs, and the asymmetry should be decided deliberately rather than absorbed by default:

- A **false pass** puts an unqualified person into a role where the harm is borne by third parties — a project, a client, a lender, a workforce — and it devalues the credential for everyone who holds it.
- A **false fail** harms one identified individual, who is usually in front of you, and who will appeal.

Because the false fail is visible and the false pass is not, assessment systems drift toward passing. The counterweight is not severity; it is written standards, evidence rules and moderation.

— 2. Evidence types, and what each can carry

The hierarchy is set out in CMP-02 §4. Here is what each type does under examination, and how each is gamed.

Evidence type	Carries	Cannot carry	Typical failure	Control
Work product authored by the candidate	Method choice, judgement in real conditions, standard of output	Whether the conditions were hard; sole authorship	Team output presented as personal	Require the reasoning to be visible, and probe it in discussion
Work product contributed to	Capability within a defined scope	Ownership of decisions made by others	Contribution overstated in retrospect	Require identifiable contribution recorded at the time
Structured professional discussion	Depth behind the artefact; reasoning under questioning	Anything the questioner did not probe	Interview skill rewarded over competence	Question from the candidate's own artefacts only; use a common question frame
Direct observation	Behaviour and interaction, which artefacts never capture	Anything outside the observed window	Performance for the observer	Observe routine work, not set-piece events
Attestation by a named referee	Corroboration of scope, duration and standing	Detail; independence from the referee's judgement	Reciprocal or reputational attestation	Require the referee's relationship, period and specific observation
Knowledge examination	Applied knowledge and reasoning against a common standard, reliably and at scale	Behaviour under real pressure	Treated as sufficient on its own	Combine with evidence types above
Self-report	A map of where to look	Capability	Uniformly favourable	Use for routing only — see §6

Two rules that resolve most disputes. **Where types conflict, the higher-ranked type wins** unless there is a stated reason to prefer the lower. And **absence of evidence is not evidence of absence** — it is absence of evidence, and the assessment record should say "not evidenced" rather than assigning a low level. The distinction matters at appeal, and it is the single most common recording error.

— 3. Writing a rubric that two people read the same way

Most rubrics fail in one of four ways, and all four are visible on the page before anyone is assessed.

Adjectival ladders. "Basic / good / strong / excellent understanding." Nothing is described; the assessor supplies the meaning, and two assessors supply different meanings.

Global impression. One overall rating for a whole competency area. It correlates with how articulate the candidate was, and it cannot be defended clause by clause at appeal.

Double-barrelled criteria. "Selects an appropriate forecasting method **and** communicates it to stakeholders." A candidate who does the first well and the second badly cannot be scored, and different assessors resolve the conflict differently.

Unbounded top levels. "Exceptional insight." Nobody can say what would satisfy it, so it is awarded on impression or never awarded at all.

The construction that works has three parts in every descriptor: **an observable action, a condition, and a standard.**

Action: selects a forecasting method — condition: where procedure does not prescribe one — standard: and records the method, the alternatives rejected and the assumptions the answer depends on.

Four supporting rules:

1. **One criterion, one judgement.** Split anything containing "and" that could be met by halves.
2. **Every level is bounded above and below.** An assessor must be able to say why this is a 3 and not a 4.
3. **Include an insufficient-evidence outcome distinct from the lowest level.** Otherwise "nothing submitted" is scored as "foundation", which is a different claim.
4. **Write the descriptors so that the difference between adjacent levels is a difference in the work, not in the adverb.** If levels 2 and 3 differ only by "consistently", the rubric will not moderate.

3.1 A worked rubric: Forecasting and EAC (PCL-AI competency 5)

Estimate at completion (EAC). Assessed from candidate-authored artefacts and a structured discussion of them.

Level	Descriptor	Evidence that satisfies it
0 — Insufficient evidence	No artefact authored or materially contributed to by the candidate, or the contribution cannot be identified.	—
1 — Foundation	Computes a forecast using the method prescribed by procedure, and identifies inputs that appear inconsistent with the project position.	A completed forecast to procedure; a query raised on an input that was subsequently corrected.
2 — Practitioner	Produces the periodic forecast for an owned scope, selecting from the methods the procedure permits, and explains the movement since the previous period in terms of its drivers.	Consecutive periodic forecasts with movement analysis; a forecast relied on without correction by the next reviewer.
3 — Professional	Selects the forecast method where procedure does not prescribe one; records the method, the alternatives rejected and the assumptions on which the answer depends; states a range where a range is warranted; and maintains the forecast under challenge, changing it on evidence and not on pressure.	A dated method-selection note; a forecast revised in a period when no external event forced it, with the basis; a documented challenge and its resolution.
4 — Leader	Sets the forecasting standard applied by others across projects — method rules, review cadence, treatment of undisclosed exposure — and can demonstrate the standard operating, including a case where it produced an unwelcome answer that was nonetheless reported.	An issued forecasting standard in use, with the candidate's authorship; back-test evidence across projects; a documented escalation route used at least once.

Note what is doing the work: level 3 is distinguished from level 2 by **who chose the method and whether the choice is recorded**, and level 4 from level 3 by **whether others apply the rule**. Neither distinction requires an assessor's opinion about quality.

— 4. Scoring: compensatory, conjunctive, and the choice between them

Compensatory scoring allows strength in one criterion to offset weakness in another — typically an average across criteria against a threshold. **Conjunctive** scoring requires a minimum on each criterion independently; no amount of excellence elsewhere compensates.

Compensatory scoring is appropriate where the competencies are genuinely substitutable in the role. Conjunctive scoring is required where a specific weakness is unacceptable regardless of other strengths — and in this framework at least two areas qualify: **responsible AI** and **human validation or human accountability**. A person who is outstanding at analysis and does not validate output is not a net positive, and an average conceals exactly that.

The practical construction most schemes should use is **conjunctive on a named subset, compensatory on the remainder**, with the subset published in advance. §8 shows how a compensatory average conceals a profile that no assessor would accept if they saw it laid out.

How PCI's own examination scores, stated plainly. It is criterion-referenced, not graded on a curve: candidates are measured against a standard, not against each other, so a strong cohort does not raise the bar and a weak one does not lower it. The examination is 90 minutes, four-option multiple choice with a single best answer, written across recall, application and analysis and weighted toward application, and results are returned immediately with a domain breakdown. The pass mark configured in the platform is **65 %**, and the definitive standard will be set by a **modified-Angoff standard-setting study** — a method in which subject-matter experts judge, item by item, the probability that a minimally competent candidate answers correctly, and the aggregate of those judgements sets the mark. **The number of items is not published, because it has not been decided**; it will be confirmed by a formal job-task analysis. EXB-05 and EXB-06 own these two studies.

— 5. Moderation: what it is for and what it finds

Moderation is not a second opinion. It is a control on the assessment process, and its outputs diagnose the instrument at least as often as the candidate.

The mechanics that make it work:

- **Independent first.** Two assessors score without seeing each other's marks. A "review" of an existing mark anchors to it and detects almost nothing.
- **A defined sample.** Full double-marking is rarely affordable. A stated proportion is moderated, plus every decision near a threshold, every fail, and every appeal [\[CONFIRM: moderation sampling rate for PCI competence assessments\]](#).
- **Reconciliation by evidence, not by rank.** Disagreements are resolved by both assessors returning to the artefact and to the rubric. Where they cannot agree, a moderator or chair rules, and the ruling is recorded with its reasoning.
- **Disagreement is analysed, not just resolved.** Every disagreement has one of four causes: the rubric is ambiguous, an assessor is miscalibrated, the portfolio is disorganised so the two assessors read different evidence, or the case is genuinely borderline. Only the last is about the candidate, and only the last requires no corrective action.

- **Calibration before the cycle.** Assessors score the same anonymised sample and discuss divergence before assessing anyone real [\[CONFIRM: assessor qualification and calibration requirements\]](#).
- **Conflict of interest declared and acted on.** An assessor does not assess a person they manage, have trained, have been employed alongside recently, or have a commercial relationship with. Separation of training from assessment is a founding principle of personnel certification and is not a formality — it is what makes the decision worth anything to a third party.
- **Records retained.** The evidence considered, the rubric applied, the levels awarded, the disagreements and their resolution. An assessment that cannot be reconstructed cannot be appealed fairly, and CER-07 — Results, scoring, appeals and complaints depends on it.

— 6. Why self-assessment alone carries no weight

This is not a claim about honesty. It follows from three structural features of the act, none of which improves with better intentions:

There is no external referent. The rater compares their work to their own experience of the work. A person whose entire exposure to forecasting has been one organisation's method rates themselves against that method, and cannot rate themselves against practice they have never seen.

The rater has an interest in the outcome. Where a self-assessment feeds a promotion, a fee or a credential, the rating is a submission in a process, and submissions are optimised.

The specific failure is invisible from inside. To know that your method selection is inadequate you must know what an adequate one looks like. A person who has never seen a method-selection note recorded does not experience its absence as a gap; they experience their own practice as normal. This is why self-assessment is least reliable exactly where it would be most useful — at the boundary between practitioner and professional.

Self-assessment nevertheless has three genuine uses, and a scheme that discards it loses them:

1. **Routing.** It tells an assessor where to look and lets a candidate assemble evidence efficiently.
2. **Structuring a professional discussion.** "You rated yourself professional here — show me the artefact" is a productive opening, and the answer is informative whichever way it goes.
3. **The gap between self and manager rating.** The disagreement is the signal. A large gap in either direction is worth an hour of conversation, and the direction tells you which hour it is: a person rating themselves well above their manager may be misreading the standard, and a person rating themselves well below may be evidencing something that is never seen.

Used for those three purposes, self-assessment is valuable. Used as a rating that stands on its own, it produces a number that reflects confidence and role identity, and it should not be recorded as an assessment outcome.

— 7. How this goes wrong

The rubric is written after the assessments. Standards are then reverse-engineered from decisions already made, which is unappealable and, when it surfaces, indefensible.

"Not evidenced" is recorded as "foundation". Two different findings. One says the candidate cannot do it; the other says nobody looked, or the candidate did not submit. Conflating them produces both unfair fails and unjustified passes, and it destroys the diagnostic value of the record.

Moderation is a signature. A moderator who reviews rather than independently scores anchors to the first mark. Agreement rates then look excellent and mean nothing.

Disagreement is resolved by seniority. The more senior assessor's view prevails without returning to the evidence. The scheme has now moved from criterion-referenced to person-referenced, and the analysis in §5 that would have found the rubric defect never happens.

A compensatory average hides an unacceptable profile. See §8. Averages are convenient and they conceal precisely the pattern a conjunctive rule exists to catch.

Assessors are never calibrated. Each is internally consistent and they differ from one another by half a level, which is invisible until a candidate assessed by the strict one appeals and cites a colleague assessed by the lenient one.

Interview performance is scored as competence. The structured discussion probes artefacts. When it becomes a general conversation, it measures fluency, which the assessment does not claim to certify and which correlates with things a fair process must not select for.

The assessment cannot be reconstructed. No record of what evidence was considered against which criterion. The first appeal exposes it, and the scheme's credibility does not recover quickly.

The examination is treated as the whole assessment. A knowledge examination is reliable, scalable and valid for what it samples. It does not sample behaviour under pressure. Scheme designers who forget this end up certifying knowledge and describing it as competence.

— 8. Worked example: a moderation reconciliation

Illustrative figures. Two assessors independently rate one candidate against five competencies on the four-level scale, coded 1 = foundation, 2 = practitioner, 3 = professional, 4 = leader.

#	Competency	Assessor A	Assessor B	Difference (A – B)
1	Cost management	3	3	0
2	Forecasting and EAC	3	2	+1
3	Predictive analytics	2	2	0
4	Project controls governance	4	3	+1
5	Human validation	2	3	–1

Agreement measures

Exact agreement: 2 criteria of 5 = 40 %. Adjacent agreement, meaning within one level: 5 of 5 = 100 %. Mean absolute difference: $(0 + 1 + 0 + 1 + 1) \div 5 = 3 \div 5 = 0.6$ levels. Mean signed difference: $(0 + 1 + 0 + 1 - 1) \div 5 = 1 \div 5 = +0.2$ levels — assessor A is running two-tenths of a level higher than assessor B on this candidate, which is a calibration observation and not yet a finding, because one candidate is not a sample.

Reconciliation, and what each disagreement diagnosed

- Criterion 2. Assessor A cited a dated method-selection note; assessor B had not seen it, because it sat in an appendix not referenced from the competency claim. Agreed at **3**. Cause: **portfolio organisation**. Action: require an evidence index mapping each artefact to the competencies it supports.
- Criterion 4. Assessor A treated authorship of a governance procedure as leader level. Assessor B applied the rubric, which requires the rule to be in use by others and a failure to have been owned; neither was evidenced. Agreed at **3**. Cause: **assessor calibration**. Action: calibration note issued to all assessors with this rubric row as the example.
- Criterion 5. Assessor B credited a validation record that assessor A read as a bare sign-off. On reading it together, the record named a date and a reviewer but not what was checked against what. Agreed at **2**. Cause: **rubric ambiguity** — the descriptor said "records the validation" without stating the minimum content. Action: rubric amended to require the independent source, the coverage and the finding.

Agreed profile: **3, 3, 2, 3, 2**.

Why the scoring rule matters. Suppose the decision rule is conjunctive on criteria 1, 2 and 4 at professional (3) or above, and practitioner (2) or above on the rest. The agreed profile meets it.

Under a compensatory rule instead — a mean of at least 2.6 — the same profile passes: $(3 + 3 + 2 + 3 + 2) \div 5 = 13 \div 5 = 2.6$. But so does the profile **4, 4, 1, 3, 1**: $(4 + 4 + 1 + 3 + 1) \div 5 = 13 \div 5 = 2.6$ — a candidate at foundation level in predictive analytics **and** in human validation, passing on the strength of two leader-level ratings elsewhere. That is the profile a conjunctive rule exists to stop, and the average made it invisible.

The finding that matters. Of three disagreements, one was about the candidate and two were about the assessment system. That ratio is normal in a young scheme, and it is the reason moderation output is reviewed by whoever owns the rubric — not filed with the candidate's result.

— 9. Checklist

Before running an assessment, and before defending one:

- [] Is every criterion written as an observable action, a condition and a standard?
- [] Does any criterion contain an "and" that could be half met?
- [] Is each level bounded above and below, so an assessor can say why not the next one up?
- [] Is there an insufficient-evidence outcome distinct from the lowest level, and is it being used?
- [] Which criteria are conjunctive, and was that published before the assessment?
- [] Do the evidence types actually sample the claim, or only the knowledge adjacent to it?
- [] Did two assessors score independently before comparing?
- [] Are moderation disagreements classified by cause — rubric, calibration, portfolio, genuine borderline?
- [] Does the moderation output go to whoever owns the rubric?
- [] Were assessors calibrated on a common sample this cycle?
- [] Are conflicts of interest declared, and is anyone assessing someone they trained or manage?
- [] Could a third party reconstruct this decision from the record in twelve months?
- [] Is any self-assessment being recorded as an outcome rather than used for routing?

— Related

- [CMP-01 – The PCI competency framework – overview](#) — what is being assessed and how each competency is written
- [CMP-02 – Competency levels and how they are evidenced](#) — the level scale and the evidence hierarchy this document applies
- [CMP-03 – PCL-AI: the fourteen competencies](#) — the source of the rubric worked in §3.1
- [CAR-07 – Building a portfolio of evidence](#) — the candidate's side of §2
- [CER-07 – Results, scoring, appeals and complaints](#) — what the assessment record has to survive
- [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — the standard-setting method referred to in §4

— Sources and standards

- PCI platform examination configuration, verified August 2026 — 90-minute duration; four-option single-best-answer items; recall, application and analysis cognitive levels, application-weighted; criterion-referenced scoring; configured pass mark 65 %; immediate result with domain breakdown. No item count exists in the platform and none is stated here.
- ISO/IEC 17024, Conformity assessment — general requirements for bodies operating certification of persons. Referenced for its principles — competence defined before assessment, impartiality, separation of training from assessment, documented and reviewable decisions. Described in our own words; no text reproduced. **PCI is not accredited** to ISO/IEC 17024 by any accreditation body.
- All figures in §8 are illustrative and recomputed from first principles.

— Status and version

Founding-stage document · Version 1.0 — effective date to be confirmed · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.